LOGARITHMIC LAPLACIAN AND OPERATORS OF LOGARITHMIC ORDER: BOUNDARY EXPANSION, POHOZAEV IDENTITY AND HADAMARD FORMULA

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Abstract. Solutions of the Dirichlet problem for the logarithmic Laplacian L_Δ with symbol $2\ln|k|$ have order at most $(\ln(1/d))^{-1/2}$ near the boundary, where d is the distance to it, and the boundary coefficient plays the role of the normal derivative. In a bounded domain of class $C^{1,1}$ every weak solution of the equation $L_\Delta u = g$ with a bounded right-hand side admits an expansion $u = \beta(\ln(1/d))^{-1/2} + O((\ln(1/d))^{-1})$ with a coefficient β continuous on the boundary. The expansion yields the Pohozaev identity, which for $N \geq 2$ had previously been formulated only under the assumption that the boundary limit exists, and the Hadamard formula for simple and multiple eigenvalues and for the torsional rigidity. For any μ, for a solution of $L_\Delta z = \mu z + f$, the identity expresses $\int_{\partial\Omega} \beta^2 (x\cdot\nu)\,\mathrm{d}\sigma$ in terms of $\|z\|^2$ and f, and the derivative of a simple eigenvalue along a deformation field X is equal to $-2\int_{\partial\Omega}\beta^2(X\cdot\nu)\,\mathrm{d}\sigma$ for a normalized eigenfunction. From this, the generic simplicity of the spectrum and, for $N\geq 2$, the strict decrease of the principal eigenvalue of a ball with an off-center hole as the hole moves away from the center are also derived. In addition, the matrix Pohozaev identity and the simplicity of the spectrum on an interval with a monotone potential are obtained. The boundary expansion, the identity and the formula carry over to operators of logarithmic order, including anisotropic ones and the logarithmic Schrödinger operator $(I-\Delta)^{\log}$. The proofs rely on a first-order equation for potentials centered on the boundary and on an exact identity comparing the solution with its dilations and translations, and require no a priori regularity of the solution.

1. Introduction

The logarithmic Laplacian L_Δ is a pseudodifferential operator with symbol $2\ln|k|$ [6]. It arises as the derivative of $(-\Delta)^s$ with respect to s at $s=0$ [6, 16] and serves as the leading part of the truncated Weil form [4]. For the Laplacian, the boundary behavior of solutions of the Dirichlet problem is described by the normal derivative, and for the fractional Laplacian $(-\Delta)^s$, by the limit of u/d^s, where d is the distance to the boundary [31]. The main result of the paper (Theorem 1.1) gives such a quantity for L_Δ: in a domain of class $C^{1,1}$ with a bounded right-hand side the limit of the product $u(x)(\ln(1/d(x)))^{1/2}$ exists at every point of the boundary and is continuous on it. All the results listed below are derived from it.

The Pohozaev identity [28] relates the solution of the Dirichlet problem for a semilinear elliptic equation to the boundary integral of the squared normal derivative. It is used to prove that in a star-shaped domain the equation with a supercritical power nonlinearity has no nontrivial solutions.

Arora, Giacomoni and Vaishnavi [1, Theorem 2.3] formulated this identity for L_Δ in a bounded domain, assuming, in particular, that the limit $u(x)^2\ln(d(x)^{-2})$ exists at the boundary. The second version of their preprint was withdrawn by the authors because of an error in the proof of the identity. In the whole $\mathbb{R}^N$, where there is no boundary term, Chen and Zhou [7, Proposition 6.1] proved the identity for nonnegative solutions of the semilinear equation.

For the fractional Laplacian the identity was obtained by Ros-Oton and Serra [31] and later extended to anisotropic integro-differential operators [32] and to pseudodifferential operators of fractional order [17]. The proof of Ros-Oton and Serra relies on the regularity of u/d^s up to the boundary [30]. For L_Δ the analogue of this regularity is the continuity near $\partial\Omega$ of the function $u(x)(\ln(1/d(x)))^{1/2}$, extended to the boundary by its limit. This continuity follows from Theorem 1.1 and the continuity of u in Ω [6, Theorem 1.11], but only a logarithmic estimate is known for its modulus of continuity. Therefore the argument of Ros-Oton and Serra does not carry over, and the identity has to be obtained differently.

We proceed to the precise statements. Let Ω be a bounded domain in $\mathbb{R}^N$, $N\geq 1$, with boundary of class $C^{1,1}$. By $d(x)$ we denote the distance from x to $\mathbb{R}^N\setminus\Omega$, by $\pi(x)$ the point of the boundary nearest

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to x (it is unique if $d(x)$ is small), and by ν the outward unit normal. By $\|\cdot\|$ we denote the norm in L^2, and by $H(\Omega)$ the space from [6] in which the weak Dirichlet problem is posed (Section 2).

Theorem 1.1 (boundary expansion)**.** *Let $u \in H(\Omega)$ be a weak solution of the problem $L_\Delta u = g$ in Ω, $u = 0$ outside Ω, with $g \in L^\infty(\Omega)$. Then $u \in L^\infty$, and for $K = \|u\|_\infty + \|g\|_\infty$ there exists a function $\beta = \beta(u) \in C(\partial\Omega)$, called the* boundary coefficient *of u, such that for almost all $x \in \Omega$ with $d(x) < r_*$*

$$\left| u(x) \Big(\ln \frac{1}{d(x)} \Big)^{1/2} - \beta(\pi(x)) \right| \le CK \Big(\ln \frac{1}{d(x)} \Big)^{-1/2}, \tag{1.1}$$

and for $x_0 \ne x_1$ in $\partial\Omega$

$$|\beta(x_0) - \beta(x_1)| \le CK \Big(\ln \frac{3D}{|x_0 - x_1|} \Big)^{-1/2}, \quad |\beta| \le CK, \tag{1.2}$$

where $D = \max(\operatorname{diam}\Omega, e)$. Here $r_* > 0$ and C depend only on N, on the radius r_0 of the two-sided ball condition for $\partial\Omega$, and on D.

The boundary coefficient is thereby uniquely determined. If continuous functions β_1 and β_2 satisfy (1.1), then $|\beta_1(\pi(x)) - \beta_2(\pi(x))| \le 2CK(\ln(1/d(x)))^{-1/2}$ for almost all x with $d(x) < r_*$, and for every $\delta > 0$ the points $\pi(x)$ of such x with $d(x) < \delta$ are dense in $\partial\Omega$, so $\beta_1 = \beta_2$. For the same reason the boundary coefficient depends linearly on the solution.

Theorem 1.2 (Pohozaev identity)**.** *Let $\mu \in \mathbb{R}$, $f \in L^\infty(\Omega) \cap W^{1,1}(\Omega)$ and $z \in H(\Omega)$ be a weak solution of the problem $L_\Delta z = \mu z + f$ in Ω, $z = 0$ outside Ω. Then z is bounded (Lemma 3.2), and, for every $a \in \mathbb{R}^N$, its boundary coefficient $\beta(z)$ from Theorem 1.1 (with $g = \mu z + f$) satisfies the equality*

$$\int_{\partial\Omega} \beta(z)^2 ((x-a)\cdot\nu)\,\mathrm{d}\sigma = \|z\|^2 + \frac{N}{2}\int_\Omega f z\,\mathrm{d}x + \int_\Omega z\,(x-a)\cdot\nabla f\,\mathrm{d}x. \tag{1.3}$$

Corollary 1.3 (eigenfunctions)**.** *Every real eigenfunction φ of L_Δ in Ω with $\|\varphi\| = 1$ satisfies, for all $a \in \mathbb{R}^N$, the equalities*

$$\int_{\partial\Omega} \beta(\varphi)^2 ((x-a)\cdot\nu)\,\mathrm{d}\sigma = 1, \quad \int_{\partial\Omega} \beta(\varphi)^2 \nu\,\mathrm{d}\sigma = 0.$$

In particular, the boundary coefficient of an eigenfunction is not identically zero, and for a radial eigenfunction of B_R it is constant and $\beta^2 = 1/(N|B_R|)$.

The boundary term in (1.3) is local: to compute it, one needs z only near the boundary, although the problem itself is nonlocal. For $N = 1$, Theorem 1.1 is close to [4, Theorem 3.8], where the solution is assumed to be bounded, but the constant in the estimate is absolute and an integral formula for β is given. Theorem 1.2 up to a factor of 2 (Section 10.1) is equivalent to Theorem 4.8 of the same work: there $f \in C(\overline{\Omega}) \cap W^{1,1}(\Omega)$, and on an interval every function of class $W^{1,1}$ has a representative continuous up to the endpoints, so the continuity condition holds automatically. In this case Corollary 1.3 gives $\beta(\varphi)^2 = 1/(2R)$ at both endpoints of the interval $(-R, R)$ for every normalized eigenfunction φ on it [4, Corollary 4.9]. For the right-hand side $\mu z + f$, formula (1.3) for $a = 0$ coincides with the identity of [1, Theorem 2.3], divided by 2.

Comparing eigenfunctions of the same operator in two nearby domains yields the Hadamard formula for $\Omega_t = \Phi_t(\Omega)$, where $\Phi_t(x) = x + tX(x)$ and $X \in C^{1,1}(\mathbb{R}^N; \mathbb{R}^N)$.

Theorem 1.4 (Hadamard formula)**.** *Let Λ be a simple eigenvalue of L_Δ in Ω and let φ be the normalized eigenfunction. Then, for small $|t|$, the operator L_Δ in Ω_t has a simple eigenvalue $\Lambda(t)$, with $\Lambda(t) \to \Lambda$ as $t \to 0$, the function $t \mapsto \Lambda(t)$ is differentiable at zero, and*

$$\Lambda'(0) = -2\int_{\partial\Omega} \beta(\varphi)^2 (X\cdot\nu)\,\mathrm{d}\sigma. \tag{1.4}$$

Corollary 1.5 (principal eigenfunction)**.** *Let $\varphi_1 > 0$ be the principal eigenfunction of L_Δ in Ω with $\|\varphi_1\| = 1$, $\beta_1 = \beta(\varphi_1)$, and let $\Lambda_1(t)$ be the principal eigenvalue of L_Δ in Ω_t. Then*

(a) *$\beta_1 > 0$ everywhere on $\partial\Omega$;*
(b) *if $X\cdot\nu \ge 0$ on $\partial\Omega$ and $X\cdot\nu \not\equiv 0$, then $\Lambda_1'(0) < 0$;*
(c) *if $\Lambda_1'(0) = 0$ for all X with $\int_{\partial\Omega} X\cdot\nu\,\mathrm{d}\sigma = 0$, then β_1 is constant on $\partial\Omega$, and for a ball the condition is satisfied and $\beta_1^2 = 1/(N|B_R|)$.*

Finally, the proofs use the precise form of the kernel only in truncation and dilation (in the anisotropic case, also its evenness and Lipschitz continuity, Section 9), and the results carry over to operators L defined on $H(\Omega)$ by the form

$$\begin{aligned} E(u,v) =& \frac{1}{2}\iint_{|x-y|<1} (u(x)-u(y))(v(x)-v(y))b(\theta)|x-y|^{-N}\,\mathrm{d}x\,\mathrm{d}y \\ &- \iint_{|x-y|\ge 1} u(x)v(y)b(\theta)|x-y|^{-N}\,\mathrm{d}x\,\mathrm{d}y \\ &+ \int Vuv\,\mathrm{d}x + \iint \kappa(x-y)u(x)v(y)\,\mathrm{d}x\,\mathrm{d}y, \quad \theta=\frac{x-y}{|x-y|}, \end{aligned} \tag{1.5}$$

where

(B1) $b: S^{N-1}\to(0,\infty)$ is even, Lipschitz and $\int_{S^{N-1}} b\,\mathrm{d}\omega = 2$;
(B2) $V\in L^\infty(\mathbb{R}^N)$;
(B3) $\kappa:\mathbb{R}^N\to\mathbb{R}$ is even and $\kappa\in L^1_{\mathrm{loc}}(\mathbb{R}^N)$.

The normalization in (B1) is achieved by multiplying the operator by a constant. The operator L_Δ corresponds to $b\equiv c_N$, $V\equiv\rho_N$, $\kappa=0$ (the constants c_N and ρ_N are defined in Section 2.2). Weak solutions of the problem $Lu=g$ and eigenfunctions are defined as in Section 2.

Theorem 1.6 (operators of logarithmic order)**.** *Suppose (B1)–(B3) hold.*

(a) *Theorem 1.1 is valid for weak solutions of problem $Lu=g$, $g\in L^\infty(\Omega)$, with constants depending also on b, $\|V\|_\infty$ and κ, and weak solutions of problem $Lu=\mu u+g$ are bounded: $\|u\|_\infty\le C_\mu(\|u\|+\|g\|_\infty)$, where C_μ is nondecreasing in $|\mu|$.*

(b) *If, moreover, $V\in W^{1,1}(U)$ for a neighborhood U of the set $\overline{\Omega}$ and $\kappa\in W^{1,1}_{\mathrm{loc}}(\mathbb{R}^N)$, then under the assumptions of Theorem 1.2 with L instead of L_Δ every weak solution of problem $Lz=\mu z+f$ satisfies the equality*

$$\begin{aligned} \int_{\partial\Omega}\beta(z)^2((x-a)\cdot\nu)\,\mathrm{d}\sigma =& \|z\|^2 + \frac{N}{2}\int_\Omega fz\,\mathrm{d}x + \int_\Omega z\,(x-a)\cdot\nabla f\,\mathrm{d}x \\ &- \frac{1}{2}\int_\Omega((x-a)\cdot\nabla V)z^2\,\mathrm{d}x \\ &- \frac{1}{2}\langle(N\kappa + y\cdot\nabla\kappa)*z, z\rangle. \end{aligned} \tag{1.6}$$

(c) *Theorems 1.4 and 8.8 are valid for L without changes.*

The main example is the logarithmic Schrödinger operator $(I-\Delta)^{\log}$ with symbol $\ln(1+|k|^2)$ [14], for which the usual derivation of the Pohozaev identity fails for lack of homogeneity under dilations.

Corollary 1.7 (logarithmic Schrödinger operator)**.** *For $L=(I-\Delta)^{\log}$, conditions (B1)–(B3) hold with $b\equiv c_N$, $V\equiv\rho_N$, and $\kappa\in W^{1,1}_{\mathrm{loc}}$, and, under the assumptions of Theorem 1.2 with L in place of L_Δ, every weak solution of the problem $Lz=\mu z+f$ satisfies*

$$\int_{\partial\Omega}\beta(z)^2((x-a)\cdot\nu)\,\mathrm{d}\sigma = \|z\|^2 - \langle G*z, z\rangle + \frac{N}{2}\int_\Omega fz\,\mathrm{d}x + \int_\Omega z\,(x-a)\cdot\nabla f\,\mathrm{d}x, \tag{1.7}$$

where G is the kernel of the operator $(I-\Delta)^{-1}$. In particular, for a normalized eigenfunction the left-hand side equals $1-\langle G*\varphi,\varphi\rangle\in(0,1)$, so that the boundary coefficient of every eigenfunction of the operator $(I-\Delta)^{\log}$ does not vanish identically.

Method. For Theorem 1.1 we test the weak equation with bump functions that shrink to a point x near the boundary, and obtain a linear first-order equation in $\xi=\ln(1/d(x))$, as in the one-dimensional case [4, Lemmas 3.6, 3.7]. It relates $u(x)$ to the potential $U_{x_0}(\xi)$, that is, the integral of $c_N u(y)|y-x_0|^{-N}$ over the exterior of the ball of radius $e^{-\xi}$ centered on the boundary point x_0. The derivative of this potential with respect to ξ is twice the average of u over the sphere $|y-x_0|=e^{-\xi}$, and at almost every interior point of this sphere the same relation holds, with the center at the boundary point nearest to that point. The main difficulty in the multidimensional case is to link different boundary points. It is overcome because the sphere of radius r sees only boundary points at a distance of at most $3r$, and potentials with such centers differ by a bounded amount. A bounded error in the potential changes the values of u, and with them the potential's derivative, only by $O(1/\xi)$, so that $U_{x_0}(\xi)(2\xi^{1/2})^{-1}$ has a limit with rate $\xi^{-1/2}$, uniformly in x_0.

For Theorem 1.2 we compare z with its dilations and translations. The equation for z is tested with the dilated function, and the equation for the dilated function with z itself, both functions being multiplied by the indicator of the intersection of the domains. The resulting cross identity is exact, the terms μz in it cancel, and it remains to find the contribution of the thin layers between the boundaries of the two domains. This contribution is found from the boundary expansion. For Theorem 1.4, the second function is an eigenfunction in the deformed domain, and the eigenvalue difference is expressed in terms of the same layers. This requires convergence of the boundary coefficients and estimates of the solutions uniform in the deformation.

What is known and what is new. The estimate $|u(x)| \le C(\ln(1/d(x)))^{-1/2}$ for bounded solutions of the Dirichlet problem with L_Δ was proved by Hernández-Santamaría, López Ríos and Saldaña [18, Theorem 1.1]. In a domain of class $C^{1,1}$, the expansion (1.1) also refines the earlier estimate $|u(x)| = O((\ln(1/d(x)))^{-\tau})$, $\tau < 1/2$, from [6, Theorem 1.11]. Hopf-type lemmas ([18, Theorem 1.4] and [29, Lemmas 3.4, 3.11]) yield a lower bound of the same order for nonnegative nontrivial supersolutions and for antisymmetric supersolutions nonnegative in a half-space. For $(I-\Delta)^{\log}$, which corresponds to the geometric 2-stable process, Kim and Mimica [20, Theorem 1.2, Example 1] obtained two-sided estimates of the Green function in bounded open sets of class $C^{1,1}$ for $N \ge 3$. The boundary Harnack principle with an explicit decay rate for nonnegative harmonic functions in any dimension is given by Theorem 1.5 of the same work. These works give the order $(\ln(1/d(x)))^{-1/2}$ of decay at the boundary for L_Δ, and for $N \ge 3$ also for $(I-\Delta)^{\log}$. In any dimension, the same order for $(I-\Delta)^{\log}$ follows from similar estimates of the Green function obtained by Cho, Kang and Kim [8, Theorem 1.7, Example 8.1]. The existence of the limit β with remainder $O((\ln(1/d(x)))^{-1})$ and the continuity of β on $\partial\Omega$ for solutions of arbitrary sign are not proved in these works. On an interval the boundary expansion and the Pohozaev identity were established in [4]. Chang-Lara and Saldaña [3] obtained Hölder-type estimates with a logarithmic modulus of continuity for L_Δ and $(I-\Delta)^{\log}$, which imply the existence of classical solutions.

The Hadamard formula for the principal eigenvalue of the fractional Laplacian is proved in [10], for any of its eigenvalues along a C^1 branch of eigenfunctions in [11, Theorem 3.1], and for nonlocal operators with integrable kernels in [2], but none of these settings covers L_Δ. To the best of our knowledge, the following results are new:

(i) the existence of the boundary coefficient for every weak solution of the Dirichlet problem for L_Δ with a bounded right-hand side in a domain of class $C^{1,1}$ in $\mathbb{R}^N$, with convergence rate (1.1) and modulus of continuity (1.2), without sign conditions (Theorem 1.1);

(ii) the Pohozaev identity for L_Δ in bounded domains of class $C^{1,1}$ in $\mathbb{R}^N$, $N \ge 2$, for every μ (Theorem 1.2);

(iii) the Hadamard formula for simple and multiple eigenvalues of L_Δ in domains of class $C^{1,1}$ in $\mathbb{R}^N$, the matrix Pohozaev identity for L_Δ and the generic simplicity of the spectrum of L_Δ and $(I-\Delta)^{\log}$ (Theorems 1.4, 8.8, 8.11, Proposition 8.9), the derivative of the torsional rigidity with respect to the domain when zero is not an eigenvalue (Theorem 8.3), and, for $N \ge 2$, the strict decrease of the principal eigenvalue of a ball with an off-center hole as the hole moves away from the center and, if $\Lambda_1(B_R) > 0$, the strict increase of the torsional rigidity of this domain (Theorem 8.6);

(iv) the boundary expansion, the Pohozaev identity and the Hadamard formula for operators of logarithmic order, including anisotropic ones and $(I-\Delta)^{\log}$ (Theorem 1.6, Corollary 1.7).

The simplicity of the spectrum of $(I-\Delta)^{\log}$ on an interval follows from [22, Theorem 1.2]. It is derived from item (iv) by another method (Corollary 9.2), which yields, moreover, a nonzero boundary coefficient of the eigenfunctions and extends to $(I-\Delta)^{\log}$ the statement on the ball and the annulus given below and the multiplicity estimate on a union of intervals (Corollary 9.4). The method of [22] is based on comparing the eigenvalues of $\psi(-\Delta)$ with the values of ψ at the eigenvalues of the Dirichlet and Neumann Laplacians and does not extend to operators $\psi(-\Delta)+V$. The Pohozaev identity of Theorem 1.6(b) yields the simplicity of the spectrum on an interval for $L_\Delta + V$ and $(I-\Delta)^{\log}+V$ with a monotone potential of class $W^{1,1}$ (Corollary 9.3). For L_Δ the simplicity on an interval is proved in [4, Corollary 4.10], and Corollary 7.6 derives it from Theorem 1.2 for $N=1$ in the same way. Moreover, in a ball and an annulus for every Λ and every spherical harmonic Y an eigenfunction of the form $\rho(|x|)Y(x/|x|)$, if it exists, is unique up to a factor (Corollaries 7.4 and 7.5).

Structure of the paper. Section 2 introduces the notation and the weak formulation, Section 3 proves the kernel truncation lemma and the boundedness of solutions, Section 4 relates the value of a solution to its potential, and Section 5 gives Theorem 1.1 and its local variant. Section 6 derives the cross identity,

and Section 7 computes the contribution of the thin layers and proves Theorem 1.2 and Corollary 1.3. Theorem 1.4, Corollary 1.5, the Hadamard formulas for torsional rigidity and for multiple eigenvalues, the monotonicity for a ball with an off-center hole, the matrix Pohozaev identity and the generic simplicity of the spectrum are obtained in Section 8. Section 9 lists the changes for operators of logarithmic order, Section 10 checks the results against the one-dimensional theory on an interval and against the radial reduction in $\mathbb{R}^3$, and Section 11 contains the conclusions.

2. Setting

2.1. **Notation.** The table lists the main notation and where each symbol is defined. Other notation is introduced where it is needed. For papers cited in an arXiv version, statement and page numbers refer to that version. Notation introduced inside a proof or a section is valid only there. If a letter means different things in different sections, this is indicated in the table.

L_Δ, E_{L}, $H(\Omega)$	the logarithmic Laplacian, its form (2.1), and the space of the weak formulation (Section 2.2)
c_N, ρ_N, γ_{E}	constants of the operator, the Euler constant (Section 2.2)
r_0, D, ν, $d(x)$, $\pi(x)$	radius of the two-sided ball condition, $D=\max(\operatorname{diam}\Omega, e)$, the outward normal, the distance to $\mathbb{R}^N\setminus\Omega$ (in [4] it is denoted by δ), and the nearest boundary point (Section 2.2)
$\langle\cdot,\cdot\rangle$, $\|\cdot\|$, $\|\cdot\|_\infty$, K	inner product and norm in L^2, norm in L^∞, $K=\|u\|_\infty+\|g\|_\infty$ (Section 2.2)
$B_r(x)$, $S_r(x)$, $\mathrm{d}\omega$, $\mathrm{d}\sigma$	ball, sphere, measures on S^{N-1} and on $\partial\Omega$ (Section 2.2)
$\ell(r)$, ξ	$\ell(r)=1/\lvert\ln\min(r,0.1)\rvert$ (Section 2.2), $\xi=\ln(1/\delta)$ for a distance or radius δ (Section 4)
$\Lambda_k(\Omega)$	eigenvalues of L_Δ in Ω counted with multiplicity (Section 2.2), in [6] they are denoted $\lambda_k^L(\Omega)$, in [4] the first of them $\lambda_1^{\mathrm{L}}(a)$
$\beta=\beta(u)$, r_*, $\beta_\pm$, β_q	boundary coefficient and width of the layer (Theorem 1.1), the boundary coefficients at the endpoints of the interval (Section 10.1 and Corollary 7.6)
$\mathcal{V}$, $\mathcal{A}$, $S(x)$	potential outside the ball of radius $d(x)/2$, weighted average of the right-hand side, essential supremum of $\lvert u\rvert$ in $B_{d(x)}(x)$ (Section 4)
j_r	kernel $c_N\lvert z\rvert^{-N}\mathbb{1}_{\lvert z\rvert\ge r}$ (Section 3)
U_{x_0}, $F(x_0,\xi)$	potential centered on the boundary and its normalization (Section 5.1)
Φ_h, T_h, J_h, Ω_h, A, $\Sigma_\pm$	dilation or translation (in Section 7.1 this is $\mathrm{id}+hX$), unitary transport of functions, the Jacobian, the image of the domain, the intersection, and the layers between the boundaries (Section 6)
X, Φ_t, Ω_t, S_t, $\mathcal{E}$, $\|\cdot\|_H$	deformation field, $\Phi_t=\mathrm{id}+tX$, deformed domain, transport of functions, the local part of the form, and the norm in $H(\Omega)$ (Section 8)
$\mathcal{W}(t)$	$\int fu_t\,\mathrm{d}x$, for $f=1$ this is the torsional rigidity (Section 8)
b, V, κ, L, $\mathcal{J}_b$	coefficients in form (1.5) of L of logarithmic order and the conditions (B1)–(B3) (Section 1), kernel $b(z/\lvert z\rvert)\lvert z\rvert^{-N}$ (Section 9.1)
j, G, G_t	kernels of the operators $(I-\Delta)^{\log}$ (Lemma 9.1) and $(I-\Delta)^{-1}$ (Corollary 1.7), Green function of the domain Ω_t (after Corollary 8.5)
l	degree of the spherical harmonic (Corollary 7.5)
q	endpoint of the interval (p,q) (Corollary 7.6)
η	$\ln(1/d(y))$ (Section 5.3), $\phi_h-\phi$ in a chart (Section 7.1, Proposition 8.9), cut-off function (Section 8.3), eigenvalues η_k of the matrix M (Theorem 8.8)
τ	$-\omega\cdot\nu(x_0)$ (Lemma 5.4), radius of the hole (Section 8.3)
ϖ	modulus of continuity of the boundary coefficient (Section 7.1)

2.2. **Operator and weak formulation.** *Operator.* Let $N\ge 1$, let $\lvert S^{N-1}\rvert$ be the area of the unit sphere, and let γ_{E} be the Euler constant. We set, as in [6, (1.5)],

$$c_N=\pi^{-N/2}\Gamma(N/2)=\frac{2}{\lvert S^{N-1}\rvert},\quad \rho_N=2\ln 2+\frac{\Gamma'(N/2)}{\Gamma(N/2)}-\gamma_{\mathrm{E}}.$$

For a smooth function u with compact support the logarithmic Laplacian is $L_\Delta u=\mathrm{d}/\,\mathrm{d}s\,(-\Delta)^s u|_{s=0}$. By [6, Theorem 1.1]

$$L_\Delta u(x)=c_N\int_{|x-y|<1}\frac{u(x)-u(y)}{|x-y|^N}\,\mathrm{d}y-c_N\int_{|x-y|\ge 1}\frac{u(y)}{|x-y|^N}\,\mathrm{d}y+\rho_N u(x),$$

and its symbol equals $2\ln|k|$. For $N=1$ we have $c_1=1$ and $\rho_1=-2\gamma_{\mathrm{E}}$.

Domain. Let $\Omega\subset\mathbb{R}^N$ be a bounded domain with boundary of class $C^{1,1}$, which is equivalent to the two-sided ball condition [24, Corollary 2]: there exists $r_0\in(0,1]$ such that for every point $x_0\in\partial\Omega$ the balls of radius r_0 with centers $x_0-r_0\nu(x_0)$ and $x_0+r_0\nu(x_0)$ lie respectively in Ω and in $\mathbb{R}^N\setminus\overline{\Omega}$, where ν is

the outward unit normal. Set $d(x) = \operatorname{dist}(x, \mathbb{R}^N \setminus \Omega)$ and $D = \max(\operatorname{diam}\Omega, e)$ and note that if $x \in \Omega$ and $d(x) < r_0$, then the point $\pi(x) \in \partial\Omega$ closest to x is unique and $x = \pi(x) - d(x)\nu(\pi(x))$. The connectedness of Ω is not used in Sections 3–7, and all statements of those sections, together with their proofs, carry over to open sets with the same properties. Below, C denotes constants depending only on N, r_0 and D, unless otherwise stated, and the value of C may change from line to line.

Weak formulation. Denote by $H(\Omega)$ the space of measurable functions $u : \mathbb{R}^N \to \mathbb{R}$ vanishing outside Ω, with $\iint_{|x-y|\le 1}(u(x)-u(y))^2|x-y|^{-N}\,\mathrm{d}x\,\mathrm{d}y < \infty$ [6, p. 3]. This is a Hilbert space embedded in $L^2(\Omega)$, and on it we define the form

$$\begin{aligned} E_{\mathrm{L}}(u,v) = {}& \frac{c_N}{2}\iint_{|x-y|<1}\frac{(u(x)-u(y))(v(x)-v(y))}{|x-y|^N}\,\mathrm{d}x\,\mathrm{d}y \\ & - c_N\iint_{|x-y|\ge 1}\frac{u(x)v(y)}{|x-y|^N}\,\mathrm{d}x\,\mathrm{d}y + \rho_N\int uv\,\mathrm{d}x. \end{aligned} \tag{2.1}$$

The Fourier transform is understood as $\hat{w}(k) = \int w(x)e^{-ik\cdot x}\,\mathrm{d}x$, and for bounded u, $v \in H(\Omega)$ the form (2.1) is written in terms of the symbol: $E_{\mathrm{L}}(u,v) = (2\pi)^{-N}\int 2\ln|k|\,\hat{u}(k)\overline{\hat{v}(k)}\,\mathrm{d}k$ (for smooth compactly supported functions [6], and by density [6, Theorem 3.1]). The passage by density is possible since both sides are continuous in the norm of $H(\Omega)$. Let $g \in L^2(\Omega)$. A function $u \in H(\Omega)$ is called a weak solution of the problem

$$L_\Delta u = g \quad (x \in \Omega), \qquad u = 0 \quad (x \in \mathbb{R}^N \setminus \Omega), \tag{2.2}$$

if $E_{\mathrm{L}}(u,\varphi) = \int g\varphi\,\mathrm{d}x$ for all $\varphi \in H(\Omega)$ [6, p. 5]. An eigenfunction φ with eigenvalue Λ is a nonzero weak solution of problem (2.2) with $g = \Lambda\varphi$, and $\Lambda_1(\Omega) < \Lambda_2(\Omega) \le \dots$ are the eigenvalues counted with multiplicity [6, Theorem 1.4] (the first inequality is strict for connected Ω).

Other notation. The positive and negative parts of a number: $s_\pm = \max(\pm s, 0)$, $\ln_+ s = \max(\ln s, 0)$. The inner product and the norm in L^2 are denoted by $\langle\cdot,\cdot\rangle$ and $\|\cdot\|$, where $\langle f, g\rangle$ also denotes $\int fg$ when the integral converges absolutely, and $\|\cdot\|_\infty$ is the norm in L^∞. For a bounded solution u of problem (2.2) set $K = \|u\|_\infty + \|g\|_\infty$, and denote by $B_r(x)$ and $S_r(x)$ the ball and the sphere of radius r with center x, $B_r = B_r(0)$, by $\mathrm{d}\omega$ the measure on S^{N-1}, and by $\mathrm{d}\sigma$ the surface measure on $\partial\Omega$ (for $N = 1$, the counting one). Solutions are defined almost everywhere, and below their values are considered at Lebesgue points. The value of a function at such a point is the limit of its averages over balls centered at it. For $r > 0$ we set $\ell(r) = 1/|\ln\min(r, 0.1)|$, as in [4]. The boundary relations below have the same form as for $N = 1$, because $c_N|S^{N-1}| = 2$, that is, the c_N-measure of a hemisphere (the measure $c_N\,\mathrm{d}\omega$ on it) equals one.

3. Kernel truncation and boundedness

The threshold $|x-y| = 1$ between the local and the far terms of the form (2.1) is arbitrary, and it can be moved to any distance r. The price is the coefficient $2\ln(1/r)$ of $\langle u, v\rangle$, and it is precisely this coefficient that produces the logarithm in the boundary behavior of solutions.

Lemma 3.1 (truncated weak form)**.** *Let u, $v \in H(\Omega)$ and $r > 0$. Then*

$$\begin{aligned} E_{\mathrm{L}}(u,v) = {}& \frac{c_N}{2}\iint_{|x-y|<r}\frac{(u(x)-u(y))(v(x)-v(y))}{|x-y|^N}\,\mathrm{d}x\,\mathrm{d}y \\ & + \Big(2\ln\frac{1}{r} + \rho_N\Big)\langle u, v\rangle - c_N\iint_{|x-y|\ge r}\frac{u(y)v(x)}{|x-y|^N}\,\mathrm{d}x\,\mathrm{d}y, \end{aligned} \tag{3.1}$$

and all three integrals converge absolutely.

Proof. For $r = 1$ this is the form (2.1). First consider $r < 1$. In the layer $r \le |x-y| < 1$ we expand the product $(u(x)-u(y))(v(x)-v(y))$. The terms $u(x)v(x)$ and $u(y)v(y)$ give

$$\frac{c_N}{2}\cdot 2\langle u, v\rangle\int_{r\le|z|<1}\frac{\mathrm{d}z}{|z|^N} = c_N|S^{N-1}|\ln\frac{1}{r}\,\langle u, v\rangle = 2\ln\frac{1}{r}\,\langle u, v\rangle,$$

and the cross terms are by symmetry equal to $-c_N\iint_{r\le|x-y|<1}u(y)v(x)|x-y|^{-N}\,\mathrm{d}x\,\mathrm{d}y$ and, together with the far term of (2.1), make up the last integral in (3.1). For $r > 1$ the same computation in the layer $1 \le |x-y| < r$, where the kernel is bounded, gives (3.1) with $\ln(1/r) < 0$. The part of the local term with $|x-y| < \min(r, 1)$ converges absolutely by the Cauchy–Bunyakovsky–Schwarz inequality for u, $v \in H(\Omega)$. Here $|z|^{-N}\mathbb{1}_{|z|\ge r}$ is bounded, and u and v lie in $L^1 \cap L^2$ and have supports in the bounded set $\overline{\Omega}$, so the far term and the layer integral converge absolutely. □

The far-field term (3.1) is $-\langle j_r * u, v\rangle$ with the kernel $j_r = c_N|z|^{-N}\mathbb{1}_{|z|\ge r} \in L^2(\mathbb{R}^N)$, and the boundedness of the solutions is deduced from Lemma 3.1 by the same device as in [16, Theorem 3.1] and [4, Theorem 3.1]: for small r the coefficient $2\ln(1/r)$ dominates the term with μ, and the far-field term is bounded.

Lemma 3.2 (boundedness)**.** *Let $\mu \in \mathbb{R}$, $g \in L^\infty(\Omega)$ and let $u \in H(\Omega)$ be a weak solution of the problem $L_\Delta u = \mu u + g$ in Ω, with $u = 0$ outside Ω. Then $\|u\|_\infty \le C(N,\mu)\|u\| + \|g\|_\infty$, where $C(N,\mu)$ can be chosen nondecreasing in μ.*

Proof. Choose $r \in (0,1)$ so that $m_r = 2\ln(1/r) + \rho_N \ge \mu + 1$, and consider the function $v = (u-t)_+$, where $t \ge 0$. Then $v \in H(\Omega)$ and $(u(x)-u(y))(v(x)-v(y)) \ge 0$. Therefore in formula (3.1) for $E_{\mathrm{L}}(u,v)$ the local term is nonnegative, and the weak equation with test function v gives

$$m_r\langle u, v\rangle \le \langle j_r * u + \mu u + g, v\rangle.$$

Here $\langle u, v\rangle = \|v\|^2 + t\int v$, since $u = v + t$ on the support of v, and $|j_r * u| \le \|j_r\|\|u\|$ by the Cauchy–Bunyakovsky–Schwarz inequality. Therefore

$$(m_r - \mu)\|v\|^2 \le [\|j_r\|\|u\| + \|g\|_\infty - (m_r - \mu)t]\int v,$$

and for $t = (\|j_r\|\|u\| + \|g\|_\infty)/(m_r - \mu)$ we obtain $v = 0$, that is, $u \le t \le \|j_r\|\|u\| + \|g\|_\infty$, since $m_r - \mu \ge 1$. For $r = \min(1/2, e^{(\rho_N - \mu - 1)/2})$ the condition $m_r \ge \mu + 1$ holds, r does not increase in μ, and for a smaller r the kernel j_r is pointwise not smaller, so the constant $C(N,\mu) = \|j_r\|$ does not decrease in μ. The same estimate for $-u$ completes the proof. □

For operators of small order, this method yields local boundedness of solutions [15, Theorem 4.1], and boundedness of eigenfunctions is proved in Theorem 1.1(1) of the same work.

4. Value of a solution and the potential

We show that the value of a bounded solution at a point x is determined up to $O(1/\ln(1/d(x)))$ by a single number, the potential of u outside the ball of radius $d(x)/2$. For $x \in \Omega$ with $\delta = d(x) < 1$ put $\xi = \ln(1/\delta)$ and

$$\mathcal{V}(x) = c_N \int_{|z-x|\ge\delta/2} u(z)|z-x|^{-N}\,\mathrm{d}z. \tag{4.1}$$

Denote by $\bar g_x(\tau)$ the average of g over the sphere $S_{(\delta/4)e^{-\tau}}(x)$, defined for almost all $\tau \ge 0$, and define $\mathcal{A}(x)$ and $S(x)$ by the equalities

$$\mathcal{A}(x) = \xi\int_0^\infty \frac{\bar g_x(\tau)}{(\xi+\tau)^2}\,\mathrm{d}\tau, \qquad S(x) = \operatorname*{ess\,sup}_{B_\delta(x)}|u|.$$

The quantity $\mathcal{A}(x)$ is a weighted average of g over the ball $B_{\delta/4}(x)$ with total mass 1, that is, $\xi\int_0^\infty(\xi+\tau)^{-2}\,\mathrm{d}\tau = 1$, and $|\mathcal{A}(x)| \le \|g\|_\infty$.

Lemma 4.1 (value and potential)**.** *Let Ω be a bounded open set, let $u \in H(\Omega)\cap L^\infty$ be a weak solution of problem (2.2) with $g \in L^\infty(\Omega)$, and let $x \in \Omega$ be a Lebesgue point of u with $d(x) < 1$.*

(a) *The following inequality holds:*

$$\left|2\ln\frac{1}{d(x)}\cdot u(x) - \mathcal{V}(x)\right| \le C_1 K. \tag{4.2}$$

(b) *If $d(x) \le 1/4$, then*

$$|2\xi u(x) - \mathcal{V}(x) - \mathcal{A}(x)| \le C_2\Big(S(x) + \frac{K}{\xi}\Big).$$

The constants C_1 and C_2 depend only on N.

Proof. We test the weak formulation with bump functions shrinking to the Lebesgue point and obtain a linear first-order equation for the corresponding averages of the solution, which is integrated explicitly.

(a) Fix $\chi_0 \in C^\infty(\mathbb{R})$, $\chi_0 \ge 0$, supported in $[1/2,1]$, with $\int_{\mathbb{R}^N}\chi_0(|y|)\,\mathrm{d}y = 1$. For $s \ge 0$ set $d_s = (\delta/4)e^{-s}$, $\varphi_s(y) = d_s^{-N}\chi_0(|y-x|/d_s)$ and $\alpha(s) = \langle u, \varphi_s\rangle$. The support of φ_s lies in $\overline{B_{d_s}(x)} \subset \Omega$. Let $\bar u(\rho)$ be the

average of u over $S_\rho(x)$ and $\tilde u(\sigma) = \bar u((\delta/4)e^{-\sigma})$. In polar coordinates

$$\alpha(s) = \int_0^{\ln 2} \tilde u(s+\theta) w(\theta)\,\mathrm{d}\theta, \quad w(\theta) = |S^{N-1}|\chi_0(e^{-\theta})e^{-N\theta} \ge 0, \quad \int w = 1,$$

and $\alpha(s) \to u(x)$ as $s \to \infty$, since x is a Lebesgue point. Apply Lemma 3.1 with $v = \varphi_s$ and $r = d_s/4$, for which $2\ln(1/r) = 2(\xi + s) + 2\ln 16$. We estimate the terms of formula (3.1).

Local term. We have $|\varphi_s(y) - \varphi_s(z)| \le \|\chi_0'\|_\infty d_s^{-N-1}|y - z|$, and pairs with $|y - z| < r$ for which at least one point lies in $B_{d_s}(x)$ are contained in $B_{2d_s}(x) \times \mathbb{R}^N$. The integral $\int_{|y-z|<r} |y - z|^{1-N}\,\mathrm{d}z$ equals $|S^{N-1}|r$, and the contribution does not exceed $C\|u\|_\infty d_s^{-N-1} d_s^N r = O(\|u\|_\infty)$.

Far term for $|z - x| \ge 2d_s$. Here $|y - z| \ge |z - x|/2$ for y in the support of φ_s, and replacing $|y - z|^{-N}$ by $|x - z|^{-N}$ introduces an error not exceeding $N2^{N+1} d_s |x - z|^{-N-1}$. Its c_N-integral over $|z - x| \ge 2d_s$ is at most $N2^{N+1}$, so the error in the far term is at most $N2^{N+1}\|u\|_\infty$. After the replacement, the part with $|z - x| \ge \delta/2$ is $\mathcal{V}(x)$, and the part with $2d_s \le |z - x| < \delta/2$ gives

$$c_N \int u(z)|x - z|^{-N}\,\mathrm{d}z = 2\int_{2d_s}^{\delta/2} \bar u(\rho)\frac{\mathrm{d}\rho}{\rho} = 2\int_{-\ln 2}^{s-\ln 2} \tilde u = 2\int_0^s \tilde u + O(\|u\|_\infty).$$

Far term for $|z - x| < 2d_s$ and $|y - z| \ge r$. Here $|y - z| < 3d_s$, and the c_N-mass of the kernel in the layer $r \le |y - z| < 3d_s$ equals $2\ln(3d_s/r) = 2\ln 12$. The contribution does not exceed $2\ln 12 \cdot \|u\|_\infty$.

Right-hand side. We have $|\langle g, \varphi_s\rangle| \le \|g\|_\infty$.

Put $M(s) = \int_0^s \alpha$. From $w \ge 0$ and $\int w = 1$ it follows that $|M(s) - \int_0^s \tilde u| \le 2\ln 2 \cdot \|u\|_\infty$. The equation $E_{\mathrm{L}}(u, \varphi_s) = \langle g, \varphi_s\rangle$ takes the form

$$(\xi + s)M'(s) - M(s) = \frac{\mathcal{V}(x)}{2} + e(s), \quad |e(s)| \le C_3 K, \quad M(0) = 0, \tag{4.3}$$

where C_3 depends only on N and χ_0. Hence $(M/(\xi + s))' = (\mathcal{V}/2 + e)/(\xi + s)^2$, and

$$\frac{M(s)}{\xi + s} = \frac{\mathcal{V}}{2}\left(\frac{1}{\xi} - \frac{1}{\xi + s}\right) + e_1(s), \quad e_1(s) = \int_0^s \frac{e(\sigma)}{(\xi + \sigma)^2}\,\mathrm{d}\sigma, \quad |e_1(s)| \le \frac{C_3 K}{\xi}.$$

Thus,

$$\alpha(s) = M'(s) = \mathcal{V}/(2\xi) + e_1(s) + e(s)/(\xi + s), \qquad |\alpha(s) - \mathcal{V}/(2\xi)| \le 2C_3 K/\xi$$

for all $s \ge 0$. As $s \to \infty$ we obtain (4.2) with $C_1 = 4C_3$.

(b) We repeat the proof of (a) with the same χ_0, d_s, φ_s, $\alpha(s)$, $\tilde u$ and $M(s)$, but estimate the errors by $S(x)$. Outside the far term the proof uses the values of u only at points of the ball $B_\delta(x)$. Namely, the local term and the part of the far term with $|z - x| < 2d_s$ use only the values of u in $B_{3d_s}(x)$, the averages $\tilde u(\sigma)$ with $\sigma \in [-\ln 2, s]$ are computed over spheres of radius at most $\delta/2$, and $|\alpha(s)| \le S(x)$. These terms and the difference $M(s) - \int_0^s \tilde u$ are at most $CS(x)$. In the far term, replacing $|y - z|^{-N}$ by $|x - z|^{-N}$ for $|z - x| \ge 2d_s$ changes it by at most $N2^{N+1} d_s c_N \int_{|z-x|\ge 2d_s} |u(z)||x - z|^{-N-1}\,\mathrm{d}z$. The part of this integral with $|z - x| < \delta$ is at most $N2^{N+1}S(x)$, and the part with $|z - x| \ge \delta$ is at most $N2^{N+2}\|u\|_\infty d_s/\delta = N2^N\|u\|_\infty e^{-s}$, since $c_N \int_{|w|>a} |w|^{-N-1}\,\mathrm{d}w = 2/a$ for $a > 0$. Equation (4.3) takes the form

$$(\xi + s)M'(s) - M(s) = \frac{\mathcal{V}(x)}{2} + \frac{1}{2}\langle g, \varphi_s\rangle + \tilde e(s), \quad |\tilde e(s)| \le C(S(x) + \|u\|_\infty e^{-s}), \quad M(0) = 0.$$

As in (a), we find $\alpha(s) = \mathcal{V}/(2\xi) + e_*(s) + e(s)/(\xi + s)$, where $e = \langle g, \varphi_s\rangle/2 + \tilde e$ is bounded and $e_*(s) = \int_0^s e(\sigma)(\xi + \sigma)^{-2}\,\mathrm{d}\sigma$, and as $s \to \infty$ we obtain

$$2\xi u(x) = \mathcal{V}(x) + \xi\int_0^\infty \frac{\langle g, \varphi_\sigma\rangle}{(\xi + \sigma)^2}\,\mathrm{d}\sigma + 2\xi\int_0^\infty \frac{\tilde e(\sigma)}{(\xi + \sigma)^2}\,\mathrm{d}\sigma.$$

The last summand does not exceed $2C(S(x) + \|u\|_\infty/\xi)$, since $\int_0^\infty e^{-\sigma}(\xi + \sigma)^{-2}\,\mathrm{d}\sigma \le \xi^{-2}$.

It remains to replace the average $\langle g, \varphi_\sigma\rangle$ by spherical averages. In polar coordinates $\langle g, \varphi_\sigma\rangle = \int_0^{\ln 2} \bar g_x(\sigma + \theta) w(\theta)\,\mathrm{d}\theta$ with the weight w from (a), and

$$\xi\int_0^\infty \frac{\langle g, \varphi_\sigma\rangle}{(\xi + \sigma)^2}\,\mathrm{d}\sigma - \mathcal{A}(x) = \xi\int_0^\infty \bar g_x(\tau)\left[\int_0^{\min(\tau, \ln 2)} \frac{w(\theta)\,\mathrm{d}\theta}{(\xi + \tau - \theta)^2} - \frac{1}{(\xi + \tau)^2}\right]\mathrm{d}\tau.$$

For $\tau \ge \ln 2$ the square bracket is at most $C(\xi + \tau)^{-3}$ in absolute value, since $\xi \ge \ln 4$, and for $\tau < \ln 2$ it is at most $C\xi^{-2}$. This bounds the difference by $C\|g\|_\infty/\xi$, which completes the proof. □

Lemma 4.1 uses neither the sign of u nor the geometry of the domain, except for the inclusion $B_{d(x)}(x) \subset \Omega$, and therefore its estimates are uniform up to the boundary. Part (b) differs from (a) in that the main contribution of the right-hand side is singled out explicitly as a quantity $\mathcal{A}(x)$ independent of the choice of the bump function, while everything else, apart from a term of order K/ξ, is estimated through the values of u in the ball $B_{d(x)}(x)$. For $N = 1$ this inequality was obtained in the proof of [4, Lemma 3.7], and our argument repeats the one-dimensional one, with the half-sum $(u(x + t) + u(x - t))/2$ replaced by the average over the sphere. Note that the error in (4.2) is of order K, whereas $2\ln(1/d)u$ and $\mathcal{V}$ grow like $(\ln(1/d))^{1/2}$ for a nonzero boundary coefficient, so relation (4.2) is meaningful right up to the boundary.

5. Boundary expansion

This section proves the boundary expansion theorem and its local variant at a single boundary point.

5.1. **Potentials centered on the boundary.** The potential (4.1) is centered at an interior point x, and to compare the values of u at different places near the boundary, we move the center to the boundary. For $x_0 \in \mathbb{R}^N$ and $\xi \in \mathbb{R}$ put

$$U_{x_0}(\xi) = c_N \int_{|y-x_0|>e^{-\xi}} u(y)|y - x_0|^{-N}\,\mathrm{d}y, \quad F(x_0, \xi) = \frac{U_{x_0}(\xi)}{2\xi^{1/2}} \quad (\xi > 0).$$

Lemma 5.1 (potentials)**.** *Let $u \in L^\infty(\mathbb{R}^N)$, $u = 0$ outside Ω, and $A \ge 1$.*

(a) *For $x_0 \in \partial\Omega$ and $\xi \ge 0$ we have $|U_{x_0}(\xi)| \le 2\|u\|_\infty(\xi + \ln D)$. Here U_{x_0} is Lipschitz, and for almost all ξ*

$$U'_{x_0}(\xi) = c_N \int_{S^{N-1}} u(x_0 + e^{-\xi}\omega)\,\mathrm{d}\omega, \quad |U'_{x_0}(\xi)| \le 2\|u\|_\infty. \tag{5.1}$$

(b) *If $|x_0 - x_1| \le Ae^{-\xi}$, then $|U_{x_0}(\xi) - U_{x_1}(\xi)| \le C(N, A)\|u\|_\infty$.*

(c) *If $x \in \Omega$, $r = d(x) < 1$ and $z \in \partial\Omega$, $|x - z| = r$, then $|\mathcal{V}(x) - U_z(\ln(1/r))| \le C(N)\|u\|_\infty$. For $x = x_0 - r\nu(x_0)$, $x_0 \in \partial\Omega$, $0 < r < r_0$, we have $d(x) = r$ and $\pi(x) = x_0$, and one can take $z = x_0$.*

(d) *If $y \notin \overline{\Omega}$, $\sigma = \operatorname{dist}(y, \Omega) < 1$ and $z \in \partial\Omega$, $|y - z| = \sigma$, then*

$$\left| c_N \int_\Omega u(x)|x - y|^{-N}\,\mathrm{d}x - U_z\Big(\ln\frac{1}{\sigma}\Big)\right| \le C(N)\|u\|_\infty.$$

For $y = x_0 + \sigma\nu(x_0)$, $x_0 \in \partial\Omega$, $0 < \sigma < r_0$, one can take $z = x_0$.

Let, in addition, $r = e^{-\xi}$, $\xi \ge \xi_A = 2\ln(4A)$ (then $4Ar \le r^{1/2}$), and $S_0(x_0) = \operatorname{ess\,sup}_{B_{\sqrt{r}}(x_0)} |u|$ for $x_0 \in \mathbb{R}^N$.

(e) *If $|x_1 - x_0| \le Ar$, then $|U_{x_0}(\xi) - U_{x_1}(\xi)| \le C(N, A)(S_0(x_0) + \|u\|_\infty r^{1/2})$.*

(f) *If $x \in \Omega$, $d(x) = r$ and $z \in \partial\Omega$, $|x - z| = r$, then $|\mathcal{V}(x) - U_z(\xi)| \le C(N)(S_0(z) + \|u\|_\infty r^{1/2})$.*

Proof. (a) Since $u = 0$ for $|y - x_0| > D$, the first estimate follows from $c_N \int_{e^{-\xi}<|y-x_0|<D} |y - x_0|^{-N}\,\mathrm{d}y = 2(\xi + \ln D)$. Formula (5.1) is obtained by passing to polar coordinates centered at x_0, and the estimate of the derivative, from $c_N|S^{N-1}| = 2$.

(b) For $r = e^{-\xi}$ and $|y - x_0| > 2Ar$ we have $|y - x_1| \ge |y - x_0|/2$ and

$$||y - x_0|^{-N} - |y - x_1|^{-N}| \le N2^{N+1}Ar|y - x_0|^{-N-1}.$$

The integral of this quantity over $|y - x_0| > 2Ar$ with weight c_N does not exceed $N2^{N+1}$, and the corresponding contribution to the difference of potentials is at most $N2^{N+1}\|u\|_\infty$. The remaining points that contribute to at least one of the two integrals lie in the layers $r < |y - x_i| < 3Ar$, $i = 0, 1$, where the c_N-mass of the kernel equals $2\ln(3A)$.

(c) For $|y - z| > 2r$ we have $|y - x| \ge |y - z|/2$ and $||y - x|^{-N} - |y - z|^{-N}| \le N2^{N+1}r|y - z|^{-N-1}$, and the c_N-integral of the right-hand side over $|y - z| > 2r$ does not exceed $N2^{N+1}$, which gives a contribution of at most $N2^{N+1}\|u\|_\infty$. The remaining points that contribute to at least one of the two integrals lie in the layers $r/2 \le |y - x| \le 3r$ and $r \le |y - z| \le 2r$ with c_N-masses of the kernel $2\ln 6$ and $2\ln 2$. If $x = x_0 - r\nu(x_0)$ with $r < r_0$, then $B_r(x)$ lies in the inner ball of radius r_0 tangent to $\partial\Omega$ at x_0, and $d(x) = r$, $\pi(x) = x_0$.

(d) Here $|x - y| \ge \sigma$ for $x \in \Omega$, and the argument is the same: the points $x \in \Omega$ with $|x - z| \le 2\sigma$ lie in the layer $\sigma \le |x - y| \le 3\sigma$. If $y = x_0 + \sigma\nu(x_0)$ with $\sigma < r_0$, then $B_\sigma(y)$ lies in the exterior ball tangent to $\partial\Omega$ at x_0, and $\operatorname{dist}(y, \Omega) = \sigma$, $z = x_0$.

(e) For $|y - x_0| > 2Ar$ we have $||y - x_0|^{-N} - |y - x_1|^{-N}| \le N2^{N+1}Ar|y - x_0|^{-N-1}$. We split this part of the integral by the sphere $|y - x_0| = r^{1/2}$ and use the equality $c_N \int_{|w|>a} |w|^{-N-1}\,\mathrm{d}w = 2/a$

from the proof of Lemma 4.1. Inside this sphere, $|u| \le S_0(x_0)$, and the contribution does not exceed $N2^{N+1}Ar \cdot S_0(x_0) \cdot 2/(2Ar) = N2^{N+1}S_0(x_0)$. Outside, the contribution does not exceed $N2^{N+1}Ar\|u\|_\infty \cdot 2r^{-1/2} = N2^{N+2}A\|u\|_\infty r^{1/2}$. The remaining points that contribute to at least one of the two integrals lie in the layers $r < |y - x_i| < 3Ar$ with c_N-mass of the kernel $2\ln(3A)$, and these layers are contained in $B_{4Ar}(x_0) \subset B_{\sqrt{r}}(x_0)$, where $|u| \le S_0(x_0)$.

(f) We argue as in (c), with the far integral split by the sphere $|y - z| = r^{1/2}$: the remaining points entering at least one of the integrals lie in $B_{3r}(z) \subset B_{\sqrt{r}}(z)$. □

Together with Lemma 5.1(c), Lemma 4.1 gives a relation between the value of the solution and the potential centered at the nearest boundary point, also independent of the geometry of Ω.

Lemma 5.2 (value, potential and nearest point)**.** *Let Ω be a bounded open set, and let $u \in H(\Omega) \cap L^\infty$ be a weak solution of problem (2.2) with $g \in L^\infty(\Omega)$. If $x \in \Omega$ is a Lebesgue point of u with $r = d(x) < 1$, and $z \in \partial\Omega$ is any point with $|x - z| = r$, then*

$$\left| 2\ln\frac{1}{r} \cdot u(x) - U_z\Big(\ln\frac{1}{r}\Big) \right| \le C_4 K, \tag{5.2}$$

where C_4 depends only on N.

Proof. It suffices to add (4.2) and the estimate of Lemma 5.1(c), since $\|u\|_\infty \le K$. □

Remark 5.3 (local constant). In Lemmas 4.1 and 5.2 the constant K can be replaced by $K_x = \|u\|_\infty + \operatorname{ess\,sup}_{B_{d(x)/4}(x)} |g|$. Indeed, in the proof of part (a) of the first lemma the right-hand side enters only through $\langle g, \varphi_s\rangle$, and the support of φ_s is contained in $\overline{B}_{\delta/4}(x)$. In part (b), the right-hand side also enters through its averages over spheres of radius at most $\delta/4$, while the second lemma relies only on part (a) of the first one.

5.2. **Spheres centered on the boundary.** By formula (5.1) the derivative of the potential is the c_N-average of u over the sphere of radius $e^{-\xi}$ centered on the boundary, so we describe how such a sphere is positioned relative to $\partial\Omega$.

Lemma 5.4 (spheres centered on the boundary)**.** *Let $x_0 \in \partial\Omega$, $0 < r < r_0/2$, $\omega \in S^{N-1}$, $\tau = -\omega \cdot \nu(x_0)$ and $y = x_0 + r\omega$. If $\tau > r/(2r_0)$, then $y \in \Omega$, and if $\tau < -r/(2r_0)$, then $y \notin \overline{\Omega}$. If $\tau \ge r/r_0$, then $r\tau/2 \le d(y) \le 3r\tau/2$ and $|\pi(y) - x_0| \le 3r$.*

Proof. We have $|y - (x_0 \mp r_0\nu(x_0))|^2 = r_0^2 + r^2 \mp 2rr_0\tau$, and the balls of radius r_0 with centers $x_0 \mp r_0\nu(x_0)$ lie in Ω and in the complement of $\overline{\Omega}$. This gives the first two statements. For $\tau \ge r/r_0$ from the inequalities $\sqrt{r_0^2 \mp \varepsilon} \le r_0 \mp \varepsilon/(2r_0)$, $0 \le \varepsilon \le r_0^2$, we obtain

$$d(y) \ge r_0 - (r_0^2 + r^2 - 2rr_0\tau)^{1/2} \ge r\tau - \frac{r^2}{2r_0} \ge \frac{r\tau}{2},$$
$$d(y) \le (r_0^2 + r^2 + 2rr_0\tau)^{1/2} - r_0 \le r\tau + \frac{r^2}{2r_0} \le \frac{3r\tau}{2}.$$

Finally, $|\pi(y) - x_0| \le d(y) + r \le 3r$. □

The sphere is divided into three parts: the exterior, where $u = 0$, a strip $|\tau| < r/r_0$ of small measure, and the regular part $\mathcal{S}_r = \{\tau \ge r/r_0\}$, where the distance to the boundary is comparable to $r\tau$. The strip has c_N-measure at most $C(N)r/r_0$, the c_N-measure of the hemisphere $\{\tau > 0\}$ equals one, and

$$E_N = c_N \int_{\tau>0} \ln\frac{1}{\tau}\,\mathrm{d}\omega < \infty$$

(for example, $E_2 = \ln 2$ and $E_3 = 1$).

5.3. **Proof of Theorem 1.1.**

Proof of Theorem 1.1. We first prove the main estimate for the derivative of the potential, then deduce the limit and the rate of convergence, and finally pass to the values of the solution and to the continuity of the boundary coefficient. By Lemma 3.2 with $\mu = 0$ the function u is bounded, and we take

$r_* = \min(r_0, 1/16)/4$ and $\xi_* = \max(\ln(1/r_*), \ln D, 2\ln 2)$. By Lemma 5.2 with $z = \pi(x)$, for almost all $x \in \Omega$ with $d(x) < r_0$ we obtain

$$\left| 2\ln\frac{1}{d(x)} \cdot u(x) - U_{\pi(x)}\Big(\ln\frac{1}{d(x)}\Big) \right| \le C_4 K. \tag{5.3}$$

Main estimate. We show that for $x_0 \in \partial\Omega$ and almost all $\xi \ge \xi_*$

$$\left| U'_{x_0}(\xi) - \frac{U_{x_0}(\xi)}{2\xi} \right| \le \frac{C_5 K}{\xi}. \tag{5.4}$$

Here $r = e^{-\xi}$ is smaller than $r_0/2$. We split the integral in (5.1) according to the parts of the sphere from Section 5.2. On the exterior part, $u = 0$, and the contribution of the strip is at most $C\|u\|_\infty r \le C\|u\|_\infty/\xi$. For $\omega \in \mathcal{S}_r$ and $y = x_0 + r\omega$, by Lemma 5.4, $\eta = \ln(1/d(y))$ satisfies $|\eta - \xi| \le \ln(1/\tau) + \ln 2$ and $\eta \ge \xi - \ln 2 \ge \xi/2$, and $y_0 = \pi(y)$ satisfies $|y_0 - x_0| \le 3r$. By (5.3) at y (for almost all ω and ξ by Fubini's theorem) $u(y) = U_{y_0}(\eta)/(2\eta) + O(K/\eta)$. Next,

$$|U_{y_0}(\eta) - U_{y_0}(\xi)| \le 2\|u\|_\infty |\eta - \xi|, \quad |U_{y_0}(\xi) - U_{x_0}(\xi)| \le C\|u\|_\infty$$

by Lemma 5.1(a) and (b) with $A = 3$, and

$$|U_{x_0}(\xi)| \left| \frac{1}{2\eta} - \frac{1}{2\xi} \right| \le \frac{2\|u\|_\infty(\xi + \ln D)|\eta - \xi|}{\xi^2} \le \frac{C\|u\|_\infty |\eta - \xi|}{\xi}.$$

Hence, on $\mathcal{S}_r$

$$u(y) = \frac{U_{x_0}(\xi)}{2\xi} + O\bigg(\frac{K(1 + \ln(1/\tau))}{\xi}\bigg).$$

Integrating this equality over $\mathcal{S}_r$ with the weight $c_N \, \mathrm{d}\omega$, we obtain (5.4), since the measure of $\mathcal{S}_r$ equals $1 - O(r)$, the integral of $\ln(1/\tau)$ is at most E_N, and $r|U_{x_0}(\xi)|/\xi \le C\|u\|_\infty r$.

It is exactly here that different boundary points are coupled: the error $O(K)$ between U_{y_0} and U_{x_0} changes U' only by $O(K/\xi)$, since u equals $U/(2\eta)$ up to an error $O(K/\eta)$. Therefore U_{x_0} satisfies the same first-order equation as in the one-dimensional case [4, Lemma 3.6], and the connection with neighboring boundary points enters only into the bounded right-hand side.

Limit. Here $F(x_0, \cdot)$ is Lipschitz on $[\xi_*, \infty)$, and by (5.4) for almost all $\xi \ge \xi_*$

$$F' = (U' - U/(2\xi))/(2\xi^{1/2}), \qquad |F'(x_0, \xi)| \le C_5 K \xi^{-3/2}.$$

Hence the limit $\beta(x_0) = \lim_{\xi\to\infty} F(x_0, \xi)$ exists and

$$|F(x_0, \xi) - \beta(x_0)| \le 2C_5 K \xi^{-1/2} \qquad (\xi \ge \xi_*) \tag{5.5}$$

uniformly in x_0, that is, $|U_{x_0}(\xi) - 2\beta(x_0)\xi^{1/2}| \le 4C_5 K$ and $2\beta(x_0) = \lim_{\xi\to\infty} \xi^{-1/2} U_{x_0}(\xi)$. By Lemma 5.1(a), this yields $|\beta| \le |F(x_0, \xi_*)| + 2C_5 K \le CK$.

Values near the boundary. If $d(x) = e^{-\xi} < r_*$ and $\xi \ge \xi_*$, then by (5.3) and (5.5)

$$\xi^{1/2} u(x) = F(\pi(x), \xi) + O(K\xi^{-1/2}) = \beta(\pi(x)) + O(K\xi^{-1/2}).$$

If, however, $\ln(1/r_*) < \xi < \xi_*$, then the left and right sides of (1.1) are bounded by CK, while $\xi^{-1/2} \ge \xi_*^{-1/2}$, and (1.1) holds with a constant depending on D.

Continuity. Let $x_0 \ne x_1$ be points of $\partial\Omega$ and $\xi = \ln(3D/|x_0 - x_1|)$. By the definition of ξ we have $|x_0 - x_1| = 3De^{-\xi}$. If $\xi \ge \xi_*$, then Lemma 5.1(b) with $A = 3D$ gives

$$|F(x_0, \xi) - F(x_1, \xi)| \le C\|u\|_\infty \xi^{-1/2},$$

which together with (5.5) leads to (1.2), and if $\xi < \xi_*$, then (1.2) follows from $|\beta| \le CK$. □

The smoothness of $\partial\Omega$ is used only through the two-sided ball condition in Lemma 5.4 and in the choice of $z = \pi(x)$ in (5.3). For $N = 1$ the sphere consists of two points ± 1, and the regular part consists of the point $\tau = 1$ corresponding to the inward direction. The estimate (5.4) is then the relation [4, Lemma 3.6(a)], in which the spherical average is replaced by the value at the point, as in Lemma 3.7 of the same work. The sign of u is never used, and the case $\beta = 0$ is not excluded.

The main estimate and the passage to the limit are local. Below they are needed to identify with β the lower limit along the normal given by a Hopf-type lemma at points where only the two-sided ball condition with a small radius holds, and we use them in the following form.

Lemma 5.5 (local main estimate)**.** *Let Ω be a bounded open set, u and g be as in Lemma 5.2, and suppose that at $z \in \partial\Omega$ the two-sided ball condition with radius $r_z \in (0,1]$ holds. Then for almost all $\xi \ge \max(\ln(2/r_z), 2\ln 2)$*

$$\left| U_z'(\xi) - \frac{U_z(\xi)}{2\xi} \right| \le C_5' K \left(\frac{1}{\xi} + \frac{e^{-\xi}}{r_z} \right),$$

where C_5' depends only on N and D. With $\xi_z = \max(\ln(2/r_z), \ln D, 2\ln 2)$, the limit $\beta(z) = \lim_{\xi\to\infty} F(z,\xi)$ exists and $|F(z,\xi) - \beta(z)| \le CK\xi^{-1/2}$ for $\xi \ge \xi_z$ with $C = C(N,D)$. If, moreover, u is continuous in Ω, then $\xi^{1/2} u(z - t\nu(z)) \to \beta(z)$ as $t \to 0+$, $\xi = \ln(1/t)$, where $\nu(z)$ is the unit vector directed from the center of the inner ball to z.

Proof. Repeat the main estimate of the proof of Theorem 1.1 at $x_0 = z$ with radius r_z instead of r_0. The statements of Lemma 5.4 about $S_r(z)$, $r = e^{-\xi}$, use only balls of radius r_z tangent to $\partial\Omega$ at z, and remain valid for $r < r_z/2$. At the points y of the regular part of the sphere, instead of (5.3) we use Lemma 5.2 with the nearest point y_0, $|y_0 - z| \le 3r$, and the estimates

$$|U_{y_0}(\eta) - U_{y_0}(\xi)| \le 2\|u\|_\infty |\eta - \xi|, \quad |U_{y_0}(\xi) - U_z(\xi)| \le C\|u\|_\infty, \quad |U_z(\xi)| \le 2\|u\|_\infty(\xi + \ln D)$$

do not depend on the geometry. The strip has c_N-measure at most $C(N)r/r_z$ and, together with the measure deficit of the regular part, contributes CKr/r_z instead of CKr. This proves the first statement. Furthermore, $F' = (U' - U/(2\xi))/(2\xi^{1/2})$ and

$$\int_\xi^\infty |F'(z,s)|\,\mathrm{d}s \le C_5' K \int_\xi^\infty \left(s^{-3/2} + \frac{e^{-s}}{r_z s^{1/2}} \right) \mathrm{d}s \le CK\xi^{-1/2} \qquad (\xi \ge \xi_z),$$

since $e^{-\xi}/r_z \le 1/2$. Finally, for $0 < t < r_z$ the inner ball tangent to the boundary at z contains $z - t\nu(z)$, and $d(z - t\nu(z)) = t$, while z is the nearest boundary point. For a continuous u all points $x \in \Omega$ are Lebesgue points, and by (5.2) $\xi^{1/2} u(z - t\nu(z)) = F(z,\xi) + O(K\xi^{-1/2})$. □

Theorem 1.1 implies an upper bound for the solution up to the boundary.

Lemma 5.6 (upper bound)**.** *If $u \in H(\Omega)$ is a weak solution of problem (2.2) with $g \in L^\infty(\Omega)$, then $|u(x)| \le C_6 K \ell(d(x))^{1/2}$ for almost all $x \in \Omega$, and for $x_0 \in \partial\Omega$, $s > 0$ the quantity $\operatorname{ess\,sup}_{B_s(x_0)} |u|$ does not exceed $C_6 K \ell(s)^{1/2}$.*

Proof. For $d(x) < r_*$ it follows from (1.1) and $|\beta| \le CK$ that $|u(x)| \le CK\xi^{-1/2} = CK\ell(d(x))^{1/2}$, since $r_* < 0.1$. For $d(x) \ge r_*$ we have $\ell(d(x)) \ge \ell(r_*)$ and $|u(x)| \le \|u\|_\infty \le \ell(r_*)^{-1/2}\|u\|_\infty \ell(d(x))^{1/2}$. The second estimate follows from the first one, since $u = 0$ outside Ω, $0 < d \le s$ on $B_s(x_0) \cap \Omega$, and ℓ is nondecreasing. □

In particular, if x_0, $z \in \partial\Omega$, then $S_0 \le C_6 K \ell(r^{1/2})^{1/2} \le 2C_6 K \xi^{-1/2}$, and the errors in Lemma 5.1(e), (f) are of order $K\xi^{-1/2}$ instead of K in (b), (c).

In a domain of class $C^{1,1}$, Lemma 5.6 gives the upper bound [18, Theorem 1.1] without barriers. In the notation of Theorem 1.1, the Hopf-type lemma of the same work (Theorem 1.4) states that $\beta > 0$ everywhere on $\partial\Omega$ for every nonnegative nonzero solution with $g \ge 0$. Such solutions are continuous [6, Theorem 1.11], and by Lemma 5.5 the lower limit from the Hopf-type lemma equals β. For eigenfunctions of arbitrary sign, Corollary 1.3 gives the weaker but unconditional assertion $\beta \not\equiv 0$.

6. Cross identity

We derive an exact relation between a solution and its dilation or translation, from which the Pohozaev identity is obtained in the next section.

Dilations and translations. Let $h > 0$ be small, and let Φ_h be either a dilation $\Phi_h(x) = (1+h)x$ or a translation $\Phi_h(x) = x + ha$ with $a \in \mathbb{R}^N$. Put $\Omega_h = \Phi_h(\Omega)$, $J_h = (1+h)^N$ for a dilation and $J_h = 1$ for a translation, and

$$(T_h w)(x) = J_h^{-1/2} w(\Phi_h^{-1} x).$$

The operator T_h maps $L^2(\Omega)$ unitarily onto $L^2(\Omega_h)$, that is, $\langle T_h u, T_h v\rangle = \langle u, v\rangle$, and $H(\Omega)$ bijectively onto $H(\Omega_h)$: under the inverse dilation the seminorm is taken over $|x - y| \le 1 + h$, and the part over $1 < |x - y| \le 1 + h$ does not exceed $C\|w\|^2$, since the kernel is bounded there.

Lemma 6.1 (scaling identity)**.** *For u, $v \in H(\Omega)$*

$$E_{\mathrm{L}}(T_h u, T_h v) = E_{\mathrm{L}}(u, v) - 2\ln(1+h)\langle u, v\rangle \tag{6.1}$$

for a dilation and $E_{\mathrm{L}}(T_h u, T_h v) = E_{\mathrm{L}}(u, v)$ for a translation. In particular, if z is a weak solution of the problem $L_\Delta z = \mu z + f$ in Ω, $z = 0$ outside Ω, then $T_h z$ is a weak solution in Ω_h of the problem

$$L_\Delta(T_h z) = \mu T_h z + T_h f - 2\ln(1+h) T_h z, \tag{6.2}$$

where the last term is absent in the case of a translation.

Proof. For a translation the identity follows from the translation invariance of the form. For a dilation with $t = 1 + h$ the substitution $x = tx'$, $y = ty'$ preserves the measure $|x-y|^{-N}\,\mathrm{d}x\,\mathrm{d}y$ up to the factor t^N, which cancels with $J_h^{-1} = t^{-N}$, and takes the condition $|x-y| < 1$ to $|x'-y'| < 1/t$. Hence $E_{\mathrm{L}}(T_h u, T_h v)$ is the right-hand side of (3.1) with $r = 1/t$ without the term $2\ln(1/r)\langle u, v\rangle = 2\ln t\langle u, v\rangle$, and Lemma 3.1 gives (6.1). This is the same as the equality of the symbols $2\ln|k/t| = 2\ln|k| - 2\ln t$. For $\varphi \in H(\Omega_h)$ the function $T_h^{-1}\varphi$ lies in $H(\Omega)$, and (6.1) with $u = z$, $v = T_h^{-1}\varphi$ gives $E_{\mathrm{L}}(T_h z, \varphi) = \langle \mu z + f - 2\ln t \cdot z, T_h^{-1}\varphi\rangle = \langle \mu T_h z + T_h f - 2\ln t \cdot T_h z, \varphi\rangle$. □

Identity (6.1) expresses the fact that L_Δ commutes with dilations up to an additive constant, and, in particular, $\Lambda_k(t\Omega) = \Lambda_k(\Omega) - 2\ln t$ for $t > 0$ and all eigenvalues (cf. [23, Lemma 2.5]). The proof of (6.1) does not depend on the sign of h.

Test functions. Let z be a weak solution of the problem $L_\Delta z = \mu z + f$ with $f \in L^\infty(\Omega)$, a solution that is bounded by Lemma 3.2. Put

$$z_1 = z, \quad z_2 = T_h z, \quad A = \Omega \cap \Omega_h, \quad \Sigma_+ = \Omega_h \setminus \Omega, \quad \Sigma_- = \Omega \setminus \Omega_h.$$

The function $z_2 \mathbb{1}_A$ lies in $H(\Omega)$, and $z_1 \mathbb{1}_A$ lies in $H(\Omega_h)$. Indeed, for bounded w, multiplication by $\mathbb{1}_A$ adds to the seminorm at most $C\int_A w^2(1+\ln_+(1/\operatorname{dist}(x,\partial A)))\,\mathrm{d}x$, and $\operatorname{dist}(x,\partial A) \ge \min(d(x), \operatorname{dist}(x, \mathbb{R}^N \setminus \Omega_h))$. Therefore $\ln(1/\operatorname{dist}(x,\partial A)) \le \ln_+(1/d(x)) + \ln_+(1/\operatorname{dist}(x, \mathbb{R}^N \setminus \Omega_h))$. Since the domains Ω and Ω_h are Lipschitz, the logarithms of the distances to their boundaries are integrable over any bounded set. If φ, $\psi \in H(B)$ for a ball $B \supset \Omega \cup \Omega_h$ and $\varphi\psi = 0$ almost everywhere, then from (2.1)

$$E_{\mathrm{L}}(\varphi, \psi) = -c_N \iint \varphi(x)\psi(y)|x-y|^{-N}\,\mathrm{d}x\,\mathrm{d}y, \tag{6.3}$$

whenever the integral on the right converges absolutely. This is the case below: the functions are bounded, $\int_A |x-y|^{-N}\,\mathrm{d}x \le C(1 + \ln_+(1/\operatorname{dist}(y, A)))$, on Σ_+ we have $\operatorname{dist}(y, A) \ge \operatorname{dist}(y, \Omega)$, on Σ_- we have $\operatorname{dist}(y, A) \ge \operatorname{dist}(y, \Omega_h)$, and it remains to use the same integrability of the logarithms of the distances.

Lemma 6.2 (cross identity)**.** *In the notation above*

$$\begin{aligned} -c_N \int_{\Sigma_+}\int_A \frac{z_2(y)z_1(x)}{|x-y|^N}\,\mathrm{d}x\,\mathrm{d}y + c_N \int_{\Sigma_-}\int_A \frac{z_1(y)z_2(x)}{|x-y|^N}\,\mathrm{d}x\,\mathrm{d}y \\ = \langle T_h f, z_1\mathbb{1}_A\rangle - \langle f, z_2\mathbb{1}_A\rangle - 2\ln(1+h)\langle z_2, z_1\mathbb{1}_A\rangle, \end{aligned} \tag{6.4}$$

where the last term is absent in the case of a translation.

Proof. Test the equation for z_1 with $z_2\mathbb{1}_A \in H(\Omega)$, and equation (6.2) for z_2 with $z_1\mathbb{1}_A \in H(\Omega_h)$:

$$\begin{aligned} E_{\mathrm{L}}(z_1, z_2\mathbb{1}_A) &= \mu\langle z_1, z_2\mathbb{1}_A\rangle + \langle f, z_2\mathbb{1}_A\rangle, \\ E_{\mathrm{L}}(z_2, z_1\mathbb{1}_A) &= \mu\langle z_2, z_1\mathbb{1}_A\rangle + \langle T_h f, z_1\mathbb{1}_A\rangle - 2\ln(1+h)\langle z_2, z_1\mathbb{1}_A\rangle. \end{aligned}$$

For a translation, the last term in the second equality is absent. Write $z_1 = z_1\mathbb{1}_A + z_1\mathbb{1}_{\Sigma_-}$ and $z_2 = z_2\mathbb{1}_A + z_2\mathbb{1}_{\Sigma_+}$. The term $E_{\mathrm{L}}(z_1\mathbb{1}_A, z_2\mathbb{1}_A)$ enters both left-hand sides, and the equal terms with μ cancel upon subtraction. The remaining terms $E_{\mathrm{L}}(z_1\mathbb{1}_{\Sigma_-}, z_2\mathbb{1}_A)$ and $E_{\mathrm{L}}(z_2\mathbb{1}_{\Sigma_+}, z_1\mathbb{1}_A)$ are computed by (6.3), and the difference of the equations gives (6.4). □

Note that the identity (6.4) is exact for every h: the test functions are multiplied by the indicator of the intersection of the domains, and no minimization is used, so there are no conditions on μ in Theorem 1.2. The left-hand side of (6.4) is concentrated in layers $\Sigma_\pm$ of thickness $O(h)$, and computing its leading term as $h \to 0$ requires z_1 and z_2 only near the boundary, which is precisely what Theorem 1.1 provides.

7. Pohozaev identity

In this section we find the contribution of the thin layers to the cross identity and derive from it the Pohozaev identity and its consequences for eigenfunctions.

7.1. **Contribution of thin layers.** The layer lemma will be needed both for the Pohozaev identity and for the Hadamard formula in Section 8, so we state it for two solutions in nearby domains. Let $X \in C^{1,1}(\mathbb{R}^N;\mathbb{R}^N)$, $\Phi_h(x) = x + hX(x)$, $\Omega_h = \Phi_h(\Omega)$, $A = \Omega \cap \Omega_h$, $\Sigma_+ = \Omega_h \setminus \Omega$, $\Sigma_- = \Omega \setminus \Omega_h$. For small h the domain Ω_h has a boundary of class $C^{1,1}$ with the radius of the two-sided ball condition bounded below uniformly in h, and Theorem 1.1 is applicable in Ω_h with constants independent of h. Indeed, $\nabla\Phi_h - I$ and the Lipschitz constant of $\nabla\Phi_h$ are $O(h)$, so in the charts the boundary of Ω_h is given by graphs of functions with $C^{1,1}$ norms bounded uniformly in h that differ from the chart functions of Ω by $O(h)$ in the C^1 norm. For $h\|\nabla X\|_\infty \le 1/20$ and $h\operatorname{Lip}(\nabla X) \le 1/4$, the image under Φ_h of each tangent ball of radius r_0 contains a tangent ball of radius $r_0/2$ at the image of the point of tangency (Taylor's formula with Lipschitz derivative), so Ω_h satisfies the two-sided ball condition of radius $r_0/2$, and $D(\Omega_h) \le 2D$. Since the ball condition of radius r_0 implies it for every smaller radius, and D enters the estimates of the same theorem only from above, its constants are common to all Ω_h. Here and to the end of Section 7.1 the constants also depend on X.

The layer Σ_+ is adjacent to the part of the boundary where $X \cdot \nu > 0$, and the layer Σ_- to the one where $X \cdot \nu < 0$, up to layers of thickness $O(h^2)$ over the set $\{|X \cdot \nu| \le Ch\}$.

Lemma 7.1 (layers). (a) *Let $\mathcal{J}$, u and v be measurable functions, $|\mathcal{J}(x-y)| \le M|x-y|^{-N}$, let u and v be bounded and vanish outside Ω and Ω_h respectively, $\|u\|_\infty + \|v\|_\infty \le K$, let α and γ be continuous on $\partial\Omega$, $|\alpha|, |\gamma| \le K$, and $\varepsilon_h \to 0$ as $h \to 0+$. Denote by d and d_h the distances to $\mathbb{R}^N \setminus \Omega$ and $\mathbb{R}^N \setminus \Omega_h$, and by δ and δ_h those to Ω and Ω_h. Suppose that for almost all $y \in \Sigma_+$ the relations hold, each with its own point $y^* \in \partial\Omega$, $|y - y^*| \le Ch$,*

$$\int_\Omega u(x)\mathcal{J}(x-y)\,\mathrm{d}x = 2\alpha(y^*)\Big(\ln\frac{1}{\delta}\Big)^{1/2} + O\Big(K + \varepsilon_h\Big(\ln\frac{1}{\delta}\Big)^{1/2} + Kh\ln\frac{1}{\delta}\Big),$$

$$v(y) = \gamma(y^*)\zeta^{-1/2} + O((\varepsilon_h + K\zeta^{-1/2})\zeta^{-1/2}), \quad \zeta = \ln\frac{1}{d_h},$$

and for almost all $y \in \Sigma_-$ the same relations hold with u and v, α and γ, Ω and Ω_h interchanged (the points y^* still lie on $\partial\Omega$), so that δ and d_h become δ_h and d. The constants in $O(\cdot)$ and ε_h do not depend on h and y. Then as $h \to 0+$

$$\int_{\Sigma_+} v(y)\int_A u(x)\mathcal{J}(x-y)\,\mathrm{d}x\,\mathrm{d}y = 2h\int_{\partial\Omega}\alpha\gamma(X\cdot\nu)_+\,\mathrm{d}\sigma + o(h),$$

$$\int_{\Sigma_-} u(y)\int_A v(x)\mathcal{J}(x-y)\,\mathrm{d}x\,\mathrm{d}y = 2h\int_{\partial\Omega}\alpha\gamma(X\cdot\nu)_-\,\mathrm{d}\sigma + o(h).$$

(b) *Let $w \in H(\Omega)$ be a weak solution of problem (2.2) with $g \in L^\infty(\Omega)$, and let $w_h \in H(\Omega_h)$ for small $h > 0$ be weak solutions of problems $L_\Delta w_h = g_h$ in Ω_h, $w_h = 0$ outside Ω_h, with $\|w\|_\infty + \|g\|_\infty + \|w_h\|_\infty + \|g_h\|_\infty \le K$ and $\beta(w_h)\circ\Phi_h \to \beta(w)$ uniformly on $\partial\Omega$. Then the assumptions of (a) hold with $u = w$, $v = w_h$, $\mathcal{J}(x-y) = c_N|x-y|^{-N}$, $\alpha = \gamma = \beta(w)$ and CK instead of K, where the conditions on u use only the equation in Ω, and those on v the problems in Ω_h and the convergence $\beta(w_h)\circ\Phi_h \to \beta(w)$. In particular, as $h \to 0+$, with $\beta = \beta(w)$,*

$$c_N\int_{\Sigma_+} w_h(y)\int_A \frac{w(x)}{|x-y|^N}\,\mathrm{d}x\,\mathrm{d}y = 2h\int_{\partial\Omega}\beta^2(X\cdot\nu)_+\,\mathrm{d}\sigma + o(h),$$

$$c_N\int_{\Sigma_-} w(y)\int_A \frac{w_h(x)}{|x-y|^N}\,\mathrm{d}x\,\mathrm{d}y = 2h\int_{\partial\Omega}\beta^2(X\cdot\nu)_-\,\mathrm{d}\sigma + o(h).$$

Proof. (a) We straighten the boundary, write the conditions in a chart, integrate over the thickness of the layer, and sum the contributions of the individual pieces. Consider the first equality. The second one is obtained in the same way by interchanging the roles of u and v, α and γ, as well as Ω and Ω_h.

Charts. Cover $\partial\Omega$ by finitely many open cylinders over balls, in each of which after a rotation $\Omega = \{x_N < \phi(x')\}$ with $\phi \in C^{1,1}$ and $|\nabla\phi| \le 1/4$, and choose a subordinate partition of unity. The cylinders cover a closed λ_0-neighborhood of $\partial\Omega$ for some $\lambda_0 > 0$. Decreasing λ_0 to the Lebesgue number of this cover, we obtain that pairs of points of this neighborhood with $|x-y| < \lambda_0$ lie in a common chart. By the implicit function theorem, in each chart $\Omega_h = \{x_N < \phi_h(x')\}$, where

$$\eta := \phi_h - \phi = hk + O(h^2), \quad k = X_N - X'\cdot\nabla\phi = (X\cdot\nu)\sqrt{1+|\nabla\phi|^2}, \quad \operatorname{Lip}(\eta) = O(h),$$

and in this chart, up to null sets, $\Sigma_+ = \{\phi < x_N < \phi_h\}$, $\Sigma_- = \{\phi_h < x_N < \phi\}$.

Replacing A by Ω. The inner integral is taken over $A = \Omega\setminus\Sigma_-$. We show that $\iint_{\Sigma_+\times\Sigma_-} |x-y|^{-N}\,\mathrm{d}x\,\mathrm{d}y = O(h^2)$, so that replacing A by Ω changes the left-hand sides only by $O(MK^2h^2)$. For $N=1$, the layers are short intervals at the endpoints, at each endpoint at most one of them is nonempty, and Σ_+ and Σ_- are separated from each other. The rest of the proof for $N=1$ reduces to a computation at each endpoint, without charts: the estimates below hold with constant ϕ and without the variable x'. For $N\ge 2$, if x and y lie in one chart, $x\in\Sigma_-$, $y\in\Sigma_+$, then on the segment between x' and y' the function η changes sign, and hence $|\eta(x')|$, $|\eta(y')| \le Ch|x'-y'|$, and $|x-y|\ge|x'-y'|$. Therefore, the contribution of such pairs does not exceed $Ch^2\iint |x'-y'|^{2-N}\,\mathrm{d}x'\,\mathrm{d}y' < \infty$, where the integral is taken over a bounded set in $\mathbb{R}^{N-1}\times\mathbb{R}^{N-1}$. Pairs that do not lie in one chart are separated by at least λ_0 and contribute at most $\lambda_0^{-N}|\Sigma_+||\Sigma_-| = O(h^2)$.

Conditions in a chart. Let $y\in\Sigma_+$, $y=(y',\phi(y')+\sigma)$, $0<\sigma<\eta(y')$, and $y_0=(y',\phi(y'))$. The distance $\delta(y)$ lies in $[c\sigma,\sigma]$, and $d_h(y)$ is comparable to $\eta(y')-\sigma$. Replacing the distance by a comparable one changes $\ln(1/\operatorname{dist})$ by $O(1)$, and hence its square root by a relative amount $O(1/\ln(1/\operatorname{dist}))$. Moreover, $|y^*-y_0|\le Ch$, and α, γ at the points y^* differ from their values at y_0 by at most $\omega(Ch)$, where ω is a common modulus of continuity of α and γ. Therefore the conditions of (a) give

$$P(y) \coloneqq \int_\Omega u(x)\mathcal{J}(x-y)\,\mathrm{d}x = 2\alpha(y_0)\Big(\ln\frac{1}{\sigma}\Big)^{1/2} + O\Big(K+\varepsilon_h'\Big(\ln\frac{1}{\sigma}\Big)^{1/2} + Kh\ln\frac{1}{\sigma}\Big)$$

and

$$v(y) = \gamma(y_0)\zeta^{-1/2} + O((\varepsilon_h' + K\zeta^{-1/2})\zeta^{-1/2}), \quad \zeta = \ln\frac{1}{\eta-\sigma},$$

where $\varepsilon_h' = \varepsilon_h + \omega(Ch)\to 0$.

Integral over the thickness. For $0<\eta\le e^{-1}$ the function

$$\Psi(\eta) = \frac{1}{\eta}\int_0^\eta \Big(\ln\frac{1}{\sigma}\Big)^{1/2}\Big(\ln\frac{1}{\eta-\sigma}\Big)^{-1/2}\,\mathrm{d}\sigma$$

tends to 1 as $\eta\to 0$, as in [4, proof of Theorem 4.8]: after the substitution $\sigma=\eta t$ the integrand tends to 1 and is dominated by $1+(\ln(1/t))^{1/2}$. The integrals of the errors from the previous part over $\sigma\in(0,\eta)$ are $O(K^2\eta(\ln(1/\eta))^{-1/2} + K\varepsilon_h'\eta + K^2h\eta(\ln(1/\eta))^{1/2})$. The last term gives the term $Kh\ln(1/\sigma)$ multiplied by $|v(y)|\le CK(\ln(1/\eta))^{-1/2}$, since $\int_0^\eta \ln(1/\sigma)\,\mathrm{d}\sigma = \eta(1+\ln(1/\eta))$, and the function $\eta(\ln(1/\eta))^{1/2}$ increases for small η and $\eta\le Ch$. Hence, the integral over the vertical segment above y' equals

$$2\alpha(y_0)\gamma(y_0)\eta_+(y')(1+o(1)) + o(h)$$

uniformly in y'.

Assembly. Multiply by a function of the partition of unity and integrate over y'. On the vertical segment of length $O(h)$ above y' the partition function differs from its value at y_0 by $O(h)$, and this gives a contribution $o(h)$. Since $\mathrm{d}y' = \mathrm{d}\sigma/\sqrt{1+|\nabla\phi|^2}$ and $\eta_+ = hk_+ + O(h^2)$, we obtain $2h\int_{\partial\Omega}\alpha\gamma(X\cdot\nu)_+\,\mathrm{d}\sigma + o(h)$.

(b) *Potential and values in the layer.* Let $y\in\Sigma_+$ and y_0, σ be as above. The point of $\partial\Omega$ closest to y is at distance at most 2σ from y_0. By Lemma 5.1(d), (b), the passage to the limit in the proof of Theorem 1.1 and the continuity of β with modulus $\varpi(\rho) = CK(\ln(3D/\rho))^{-1/2}$ from (1.2)

$$c_N\int_\Omega w(x)|x-y|^{-N}\,\mathrm{d}x = 2\beta(y_0)\Big(\ln\frac{1}{\sigma}\Big)^{1/2} + O(K),$$

and since $\delta(y)\in[c\sigma,\sigma]$, this is the first condition of (a) with $y^*=y_0$. The point of $\partial\Omega_h$ closest to y is $\Phi_h(x_1)$ with $|x_1-y_0|\le Ch$. By the boundary expansion (1.1) in Ω_h

$$w_h(y) = \beta(w_h)(\Phi_h(x_1))\zeta^{-1/2} + O(K\zeta^{-1}), \quad \zeta = \ln\frac{1}{d_h(y)},$$

and $|\beta(w_h)(\Phi_h(x_1)) - \beta(x_1)|\le\varepsilon_h$, where $\varepsilon_h = \sup_{\partial\Omega}|\beta(w_h)\circ\Phi_h - \beta| \to 0$. This is the second condition of (a) with $y^*=x_1$. In Σ_- the values of w are given by (1.1) in Ω with $y^*=\pi(y)$, and the potential of w_h by item (d) of the same lemma in Ω_h with $y^*=x_2$, where $\Phi_h(x_2)$ is the point of $\partial\Omega_h$ closest to y, so that $|y-x_2|\le Ch$. Replacing $\beta(w_h)$ by β gives an error $\varepsilon_h(\ln(1/\delta_h))^{1/2}$. By (a) we obtain the equalities of (b). □

Below, Lemma 7.1 is applied to $w=z$ and $w_h=z_2=T_hz$, where $X(x)=x$ for dilation and $X(x)=a$ for translation. The proof of the lemma uses only the values of X in a neighborhood of $\overline{\Omega}$, so the field can be cut off outside it. Here T_hz solves the problem (6.2) with a bounded right-hand side, and $\beta(T_hz)(\Phi_h(x_0)) = J_h^{-1/2}\beta(z)(x_0)$, since under dilation the distance to the boundary is multiplied by $1+h$,

while $\ln(1/((1+h)d))/\ln(1/d) \to 1$ as $d \to 0$. Under a translation, $J_h = 1$ and distances do not change. Since $J_h \to 1$, the assumptions of the lemma are satisfied.

Lemma 7.2 (right-hand side)**.** *Let z, z_1, z_2, A and T_h be as in Section 6, with $f \in L^\infty(\Omega) \cap W^{1,1}(\Omega)$. As $h \to 0+$*

$$\frac{1}{h}(\langle T_h f, z_1 \mathbb{1}_A\rangle - \langle f, z_2 \mathbb{1}_A\rangle) \to -N\langle f, z\rangle - 2\langle z, x\cdot\nabla f\rangle$$

for dilation, while for translation the limit equals $-2\langle z, a\cdot\nabla f\rangle$. Moreover, $h^{-1}\cdot 2\ln(1+h)\langle z_2, z_1\mathbb{1}_A\rangle \to 2\|z\|^2$.

Proof. Consider a dilation, $t = 1+h$, $J_h = t^N$. For a translation, everything is repeated with $\Phi_h^{\pm1}x = x \pm ha$ and without the factors $J_h^{\mp1/2}$. By the change of variables $x = ty$ in the second inner product

$$\langle T_h f, z\mathbb{1}_A\rangle - \langle f, (T_h z)\mathbb{1}_A\rangle = h\int_\Omega z(y)q_h(y)\,\mathrm{d}y, \quad q_h(y) = \frac{J_h^{-1/2}\mathbb{1}_A(y)f(y/t) - J_h^{1/2}\mathbb{1}_A(ty)f(ty)}{h}.$$

If $d(y) > Ch$, then the points $e^r y$, $|r| \le \ln t$, lie in Ω, since $|e^r y - y| \le Ch$. Therefore both indicator functions equal one and

$$q_h(y) = \frac{J_h^{-1/2}(f(y/t) - f(y)) - J_h^{1/2}(f(ty) - f(y)) + (J_h^{-1/2} - J_h^{1/2})f(y)}{h}.$$

For each $\delta > 0$ the difference quotients $(f(e^{\pm\sigma}y) - f(y))/\sigma$ as $\sigma \to 0+$ converge to $\pm y\cdot\nabla f$ in $L^1(\{d > \delta\})$, since the segments between y and $e^{\pm\sigma}y$ lie in Ω, and the derivative $J_h^{\mp1/2}$ at $h = 0$ equals $\mp N/2$. Hence $q_h \to -Nf - 2x\cdot\nabla f$ in $L^1(\{d > \delta\})$. The remaining points form a layer $\{d \le Ch\}$ of measure $O(h)$, where $|q_h| \le C\|f\|_\infty/h$, so the integrals $\int_\Omega |q_h|$ are bounded. We split $\int zq_h$ into parts over $\{d \le \delta\}$ and $\{d > \delta\}$. The first does not exceed $CK\ell(\delta)^{1/2}\sup_h \int |q_h|$, since $|z| \le CK\ell(\delta)^{1/2}$ on $\{d \le \delta\}$ by Lemma 5.6. The second converges to $\int_{\{d>\delta\}} z(-Nf - 2x\cdot\nabla f)$, since z is bounded. Letting $\delta \to 0$, we obtain the first assertion. The last assertion follows from $T_h z \to z$ and $z\mathbb{1}_A \to z$ in L^2. □

7.2. **Proofs of Theorem 1.2 and Corollary 1.3.**

Proof of Theorem 1.2. By Lemma 3.2, z is bounded, and Theorem 1.1 is applicable to z with $g = \mu z + f$. Divide (6.4) by h and pass to the limit $h \to 0+$ by Lemmas 7.1 and 7.2. For the dilation

$$-2\int_{\partial\Omega}\beta^2(x\cdot\nu)_+\,\mathrm{d}\sigma + 2\int_{\partial\Omega}\beta^2(x\cdot\nu)_-\,\mathrm{d}\sigma = -N\langle f, z\rangle - 2\langle z, x\cdot\nabla f\rangle - 2\|z\|^2,$$

that is, (1.3) for $a = 0$. For the translation, $\int_{\partial\Omega}\beta^2(a\cdot\nu)\,\mathrm{d}\sigma = \langle z, a\cdot\nabla f\rangle$, and subtraction yields (1.3) for every a. □

In particular, Theorem 1.2 gives two identities:

$$\int_{\partial\Omega}\beta^2(x\cdot\nu)\,\mathrm{d}\sigma = \|z\|^2 + \frac{N}{2}\langle f, z\rangle + \langle z, x\cdot\nabla f\rangle, \quad \int_{\partial\Omega}\beta^2\nu\,\mathrm{d}\sigma = \int_\Omega z\nabla f\,\mathrm{d}x. \tag{7.1}$$

The first corresponds to dilation, the second, to translation. Note that the proof uses only the limits as $h \to 0+$, and no regularity of z beyond boundedness is needed.

Proof of Corollary 1.3. Apply Theorem 1.2 with $\mu = \Lambda$ and $f = 0$. Then the right-hand side of (1.3) equals $\|\varphi\|^2 = 1$ for every a, while the second identity of (7.1) yields $\int\beta^2\nu\,\mathrm{d}\sigma = 0$. If $\beta \equiv 0$, the left-hand side of (1.3) would equal zero rather than one. For a radial eigenfunction of the ball B_R the boundary coefficient is constant: for an orthogonal transformation $\mathcal{O}$ fixing the center of the ball, $d\circ\mathcal{O} = d$ and $\pi\circ\mathcal{O} = \mathcal{O}\circ\pi$, so the function $\varphi\circ\mathcal{O} = \varphi$ has boundary coefficient $\beta\circ\mathcal{O}$. Since the boundary coefficient is unique by (1.1), $\beta\circ\mathcal{O} = \beta$. Such transformations act transitively on the sphere ∂B_R (for $N = 1$ this is the reflection $x \mapsto -x$). Since $x\cdot\nu = R$ on ∂B_R, we obtain $\beta^2 R|\partial B_R| = 1$, that is, $\beta^2 = 1/(N|B_R|)$. □

For every normalized eigenfunction of B_R, Corollary 1.3 with $a = 0$ gives $\int_{\partial B_R}\beta^2\,\mathrm{d}\sigma = 1/R$, that is, the average of β^2 over the sphere equals $1/(N|B_R|)$.

Lemma 7.3 (the principal eigenfunction of the ball)**.** *The principal eigenfunction of the ball B_R is radial, and its boundary coefficient is constant.*

Proof. The principal eigenvalue is simple, and the principal eigenfunction is positive [6, Theorem 1.4(iii)]. After an orthogonal transformation fixing the center of the ball it remains a positive normalized eigenfunction of this eigenvalue and, by simplicity, coincides with the original one. It remains to apply the last statement of Corollary 1.3. □

In the general case, when $M_a = \max_{\partial\Omega}(x-a)\cdot\nu > 0$ (the integral of $(x-a)\cdot\nu$ over the boundary equals $N|\Omega|$), Corollary 1.3 gives, for every normalized eigenfunction,

$$1 = \int_{\partial\Omega} \beta^2 (x-a)\cdot\nu \,\mathrm{d}\sigma \le M_a \int_{\partial\Omega} \beta^2 \,\mathrm{d}\sigma.$$

Equality here holds if and only if $\beta = 0$ on the set $\{(x-a)\cdot\nu < M_a\}$, for example for all eigenfunctions of a ball centered at a. If Ω is convex and contains $B_r(a)$, then on its boundary $(x-a)\cdot\nu \ge r$, and hence $1/M_a \le \int_{\partial\Omega}\beta^2\,\mathrm{d}\sigma \le 1/r$ for every normalized eigenfunction.

Corollary 7.4 (radial eigenfunctions of the ball)**.** *Every eigenvalue of L_Δ in the ball B_R has, up to a factor, at most one radial eigenfunction.*

Proof. Let φ_1, φ_2 be orthogonal normalized real radial eigenfunctions with the same eigenvalue. Their boundary coefficients are constant, $\alpha_i = \beta(\varphi_i)$, and by Corollary 1.3 $\alpha_i^2 = 1/(N|B_R|) \ne 0$. Here $\psi = \alpha_2\varphi_1 - \alpha_1\varphi_2$ is a radial eigenfunction of this eigenvalue with $\|\psi\|^2 = \alpha_1^2 + \alpha_2^2 > 0$ and $\beta(\psi) = 0$ (the map $u \mapsto \beta(u)$ is linear by the uniqueness of the coefficient in (1.1)), which contradicts the same corollary. □

The same argument applies to every spherical harmonic, as well as in an annulus and on unions of annuli. By $\mathcal{H}_l$ we denote the space of real spherical harmonics of degree l on S^{N-1}.

Corollary 7.5 (one spherical harmonic)**.** *Let $N \ge 2$, let Ω be the ball B_R or the annulus $\{R_1 < |x| < R_2\}$, and let $Y \in \mathcal{H}_l$, $Y \ne 0$. Then for each eigenvalue Λ the eigenfunctions of the form $\rho(|x|)Y(x/|x|)$ together with zero make up a subspace of dimension at most one. If Ω is the union of n annuli centered at the origin or of the ball B_R and $n-1$ such annuli with pairwise disjoint closures, then the dimension of this subspace is at most n.*

Proof. Theorem 1.2 with $\mu = \Lambda$, $f = 0$ and $a = 0$ gives for every real eigenfunction φ the equality $\int_{\partial\Omega}\beta(\varphi)^2(x\cdot\nu)\,\mathrm{d}\sigma = \|\varphi\|^2$, where $x\cdot\nu = r$ on the outer boundary sphere of radius r and $x\cdot\nu = -r$ on the inner one. If the boundary coefficient of the eigenfunction vanishes on all outer spheres, the left-hand side is nonpositive, and $\varphi = 0$. Let $\varphi = \rho(|x|)Y(x/|x|)$, and let S_r be a boundary sphere. By Fubini's theorem, for almost every $t \ne r$ close to r and such that $tS^{N-1} \subset \Omega$, the inequality (1.1) at $t\omega$ holds for almost all ω, that is, $|c(t)Y(\omega) - \beta(r\omega)| \le CK(\ln(1/|r-t|))^{-1/2}$ with $c(t) = \rho(t)(\ln(1/|r-t|))^{1/2}$, and by continuity of both sides it holds for every ω. Hence $c(t)$ converges to some b_r depending linearly on φ, and $\beta(\varphi)(r\omega) = b_r Y(\omega)$. The map $\varphi \mapsto (b_r)$, where r runs over the radii of the outer spheres, is injective on the subspace in question, so the dimension of this subspace is at most the number of such radii. □

Corollary 7.6 (simplicity of the spectrum on an interval)**.** *All eigenvalues of the Dirichlet problem for L_Δ on an interval are simple.*

Proof. This is [4, Corollary 4.10]. We include the proof. Let $\Omega = (p,q)$. Corollary 1.3 with $a = p$ gives $(q-p)\beta_q(\varphi)^2 = 1$ for every normalized real eigenfunction φ. If φ_1, φ_2 are orthogonal normalized eigenfunctions with the same eigenvalue and $\alpha_i = \beta_q(\varphi_i)$, then $\alpha_i \ne 0$, and $\psi = \alpha_2\varphi_1 - \alpha_1\varphi_2$ has $\|\psi\|^2 = \alpha_1^2 + \alpha_2^2 > 0$ and $\beta_q(\psi) = 0$, which contradicts the same equality for $\psi/\|\psi\|$. □

For the fractional Laplacian on an interval this argument was given by Kwaśnicki [21], and the result itself was obtained by Fall, Ghimenti, Micheletti and Pistoia [13] by a different approach. For radial eigenfunctions of the fractional Laplacian in a ball, an annulus and on unions of annuli the same argument was used in [11, Proposition 4.3, Theorems 4.1, 4.2], and for $l = 0$, Corollary 7.5 is its analogue.

Corollary 7.7 (union of intervals)**.** *On a union of n intervals (p_i, q_i) with pairwise disjoint closures the multiplicity of each eigenvalue of L_Δ is at most n.*

Proof. For $a = \max_i q_i$ Theorem 1.2 with $f = 0$ gives for the eigenfunction φ

$$\sum_i (a-p_i)\beta_{p_i}(\varphi)^2 - \sum_i (a-q_i)\beta_{q_i}(\varphi)^2 = \|\varphi\|^2,$$

and the map $\varphi \mapsto (\beta_{p_i}(\varphi))_i$ is injective on the eigenspace: if all $\beta_{p_i}(\varphi) = 0$, then by this identity $\varphi = 0$. □

On a symmetric interval every eigenfunction is even or odd: the reflection carries it into an eigenfunction of this eigenvalue, and it is simple.

8. Hadamard formula

In this section the sets $A = \Omega \cap \Omega_t$ and $\Sigma_\pm$ are constructed from Ω_t in the same way as in Section 6 from Ω_h, J_t is the Jacobian of Φ_t, and the constants depend also on X and Λ.

Spectral inequalities and the asymptotics of the eigenvalues of the operator $\frac{1}{2}\log(-\Delta) = L_\Delta/2$ were obtained by Laptev and Weth [23], while here the dependence of the eigenvalues on the domain is studied. The cross identity of Section 6 does not require the second function to be obtained from the first one by a dilation or a translation. If one takes eigenfunctions of the same operator in two nearby domains, then the difference of the equations gives an exact expression for the difference of the eigenvalues through the same thin layers, and Lemma 7.1 turns it into the Hadamard formula. Let $X \in C^{1,1}(\mathbb{R}^N;\mathbb{R}^N)$ (only the values of X in a neighborhood of $\overline{\Omega}$ matter, and outside it X can be multiplied by a cutoff), $\Phi_t(x) = x + tX(x)$ and $\Omega_t = \Phi_t(\Omega)$. For small $|t|$ the mapping Φ_t is a diffeomorphism of class $C^{1,1}$, Ω_t is a domain of class $C^{1,1}$ with r_0 bounded below and D bounded above uniformly in t, and the operator $S_t u = u \circ \Phi_t^{-1}$ maps $H(\Omega)$ bijectively onto $H(\Omega_t)$: the inverse operator $v \mapsto v \circ \Phi_t$ has the same form. By $\|u\|_H^2 = \mathcal{E}(u,u) + \|u\|^2$ we denote the squared norm in $H(\Omega)$, where $\mathcal{E}(u,u) = (c_N/2)\iint_{|x-y|<1}(u(x)-u(y))^2|x-y|^{-N}\,\mathrm{d}x\,\mathrm{d}y$.

8.1. Simple eigenvalue.

First we compare the forms in Ω and in Ω_t.

Lemma 8.1 (transport of the form)**.** *For u, $v \in H(\Omega)$ and small $|t|$*

$$|E_{\mathrm{L}}(S_t u, S_t v) - E_{\mathrm{L}}(u,v)| \le C|t|\|u\|_H\|v\|_H, \quad |\langle S_t u, S_t v\rangle - \langle u, v\rangle| \le C|t|\|u\|\|v\|,$$

where C depends on N, D and $\|X\|_{C^1}$.

Proof. Change the variables $x = \Phi_t(x')$, $y = \Phi_t(y')$ and denote by J_t the Jacobian of Φ_t. Since $|\Phi_t(x') - \Phi_t(y') - (x'-y')| \le |t|\|\nabla X\|_\infty |x'-y'|$ and $J_t = 1 + O(t)$, the weight $|x'-y'|^N |\Phi_t(x') - \Phi_t(y')|^{-N} J_t(x') J_t(y')$ equals $1 + O(t)$. Therefore, the local terms of the form (2.1) on the pairs $S_t u$, $S_t v$ and u, v differ by $O(t)\mathcal{E}(u,u)^{1/2}\mathcal{E}(v,v)^{1/2}$ and by an integral over the layer $1 - C|t| \le |x'-y'| \le 1 + C|t|$. The distance between points in the layer is close to one, and by Young's inequality this integral is at most $C|t|\|u\|\|v\|$. In the far term the kernel is bounded, and its change by $O(t)$ gives $C|t|\|u\|_{L^1}\|v\|_{L^1} \le C|t|\|u\|\|v\|$, while the integral over this layer, as above, is at most $C|t|\|u\|\|v\|$ by Young's inequality. The term with ρ_N changes by $\rho_N(\langle S_t u, S_t v\rangle - \langle u, v\rangle)$. Finally, $\langle S_t u, S_t v\rangle = \int uv J_t \,\mathrm{d}x'$. □

Lemma 8.2 (continuity of a simple eigenvalue)**.** *Let Λ be a simple eigenvalue of L_Δ in Ω and φ the normalized eigenfunction. Then for small $|t|$ the operator L_Δ in Ω_t has exactly one eigenvalue $\Lambda(t)$ near Λ. This eigenvalue is simple, and $|\Lambda(t) - \Lambda| \le C|t|$. Normalized eigenfunctions φ_t can be chosen so that $\varphi_t \circ \Phi_t \to \varphi$ in $L^2(\Omega)$, $\varphi_t \to \varphi$ in $L^1(\mathbb{R}^N)$, $\|\varphi_t\|_\infty \le C$ and $\beta(\varphi_t) \circ \Phi_t \to \beta(\varphi)$ uniformly on $\partial\Omega$.*

Proof. Eigenvalues. The spectrum of L_Δ in Ω_t consists of the minimax values of $E_{\mathrm{L}}(S_t u, S_t u)/\|S_t u\|^2$ on $H(\Omega)$. We have $E_{\mathrm{L}}(u,u) + C\|u\|^2 \ge c\|u\|_H^2$, since the far term and the term with ρ_N are bounded in L^2, with the constants C and c common to all Ω_t with $|\Omega_t| \le 2|\Omega|$. Therefore Lemma 8.1, applied to the spans of the first k eigenfunctions in Ω and in Ω_t, where $\|u\|_H \le C_k\|u\|$, together with $\|S_t u\|^2 = (1 + O(t))\|u\|^2$ gives $|\Lambda_k(\Omega_t) - \Lambda_k(\Omega)| \le C_k|t|$ for each k, and the simplicity of Λ is preserved for small $|t|$. The estimate $\|u\|_H \le C_k\|u\|$ for eigenfunctions in Ω_t follows from the lower bound for the form, since $\Lambda_k(\Omega_t)$ are bounded above uniformly in t: this is given by the same lemma, applied to eigenfunctions in Ω.

Eigenfunctions. Set $\psi_t = \varphi_t \circ \Phi_t$ and choose the sign of φ_t so that $\langle \psi_t, \varphi\rangle \ge 0$. The functions ψ_t are bounded in $H(\Omega)$: by Lemma 8.1, $E_{\mathrm{L}}(\psi_t, \psi_t) \le \Lambda(t) + C|t|\|\psi_t\|_H^2$, for small $|t|$ the last term is absorbed by the estimate $E_{\mathrm{L}}(u,u) + C\|u\|^2 \ge c\|u\|_H^2$, and $\|\psi_t\|^2 = 1 + O(t)$. For any sequence $t_n \to 0$ there is a subsequence converging weakly in $H(\Omega)$ and, by the compactness of the embedding [6, (3.3)], strongly in L^2 to a function ψ with $\|\psi\| = 1$. Passing to the limit in $E_{\mathrm{L}}(S_t\psi_t, S_t v) = \Lambda(t)\langle S_t\psi_t, S_t v\rangle$ for a fixed $v \in H(\Omega)$ by means of the same lemma, we obtain $E_{\mathrm{L}}(\psi, v) = \Lambda\langle\psi, v\rangle$, that is, $\psi = \pm\varphi$, and the sign condition gives $\psi = \varphi$. Hence $\psi_t \to \varphi$ in L^2, and therefore $\varphi_t \to \varphi$ in $L^1(\mathbb{R}^N)$, since the Jacobian of Φ_t is bounded, $\|(\psi_t - \varphi)\circ\Phi_t^{-1}\|_{L^1} \le C\|\psi_t - \varphi\|_{L^1}$ and $\varphi \circ \Phi_t^{-1} \to \varphi$ in L^1. By Lemma 3.2 with $\mu = \Lambda(t)$, $\|\varphi_t\|_\infty \le C$.

Boundary coefficients. Fix $x_0 \in \partial\Omega$ and $\xi \ge \xi_*$, where ξ_* is from the proof of Theorem 1.1. By (5.5), applied in Ω and in Ω_t with common constants,

$$|\beta(\varphi_t)(\Phi_t(x_0)) - \beta(\varphi)(x_0)| \le |F_t(\Phi_t(x_0),\xi) - F(x_0,\xi)| + C\xi^{-1/2},$$

where F_t is constructed from φ_t. For a fixed ξ the kernel $\mathbb{1}_{|y-x|>e^{-\xi}}|y-x|^{-N}$ is bounded by $e^{N\xi}$ and depends continuously on x in L^1_{loc}. Hence the first term does not exceed the sum of $Ce^{N\xi}\|\varphi_t - \varphi\|_{L^1}$ and a term that, for the same ξ, tends to zero together with $|t|\|X\|_\infty$. Choosing first ξ and then t, we obtain uniform convergence. □

Proof of Theorem 1.4. As in the derivation of the Pohozaev identity, we compare the equations in two domains and find the contribution of the thin layers. Here $\varphi_t \mathbb{1}_A$ lies in $H(\Omega)$, and $\varphi \mathbb{1}_A$, in $H(\Omega_t)$, as in Section 6. Test the equation for φ with $\varphi_t \mathbb{1}_A$ and the equation for φ_t with $\varphi \mathbb{1}_A$, and subtract:

$$E_{\mathrm{L}}(\varphi_t, \varphi\mathbb{1}_A) - E_{\mathrm{L}}(\varphi, \varphi_t\mathbb{1}_A) = (\Lambda(t) - \Lambda)\langle \varphi_t, \varphi\mathbb{1}_A\rangle.$$

Write $\varphi_t = \varphi_t\mathbb{1}_A + \varphi_t\mathbb{1}_{\Sigma_+}$ and $\varphi = \varphi\mathbb{1}_A + \varphi\mathbb{1}_{\Sigma_-}$. The symmetric terms cancel, and by (6.3)

$$(\Lambda(t) - \Lambda)\langle \varphi_t, \varphi\mathbb{1}_A\rangle = -c_N \int_{\Sigma_+}\int_A \frac{\varphi_t(y)\varphi(x)}{|x-y|^N}\,\mathrm{d}x\,\mathrm{d}y + c_N \int_{\Sigma_-}\int_A \frac{\varphi(y)\varphi_t(x)}{|x-y|^N}\,\mathrm{d}x\,\mathrm{d}y. \tag{8.1}$$

This equality is exact for each t. For $t > 0$ the assumptions of Lemma 7.1(b) with $w = \varphi$ and $w_t = \varphi_t$ hold by Lemma 8.2, and the right-hand side equals

$$-2t \int \beta^2 (X\cdot\nu)_+ + 2t \int \beta^2 (X\cdot\nu)_- + o(t) = -2t \int \beta^2 (X\cdot\nu) + o(t).$$

Moreover, $\langle \varphi_t, \varphi\mathbb{1}_A\rangle \to 1$, since $\varphi_t \to \varphi$ in L^1 with $\|\varphi_t\|_\infty \le C$, while $\varphi\mathbb{1}_A \to \varphi$ in L^2. For $t < 0$ we have $\Phi_t = \mathrm{id} + |t|(-X)$, and the argument with $-X$ gives the same limit of the ratio $(\Lambda(t) - \Lambda)/t$. □

For the dilation $X(x) = x$, formula (1.4) together with Corollary 1.3 gives $\Lambda'(0) = -2$, which coincides with the derivative of the exact equality $\Lambda(\Omega_t) = \Lambda - 2\ln(1+t)$ (Section 6), while for a translation it gives $\Lambda'(0) = 0$, in agreement with $\int \beta^2 \nu\,\mathrm{d}\sigma = 0$. Conversely, for simple eigenvalues the same corollary can be derived from the Hadamard formula, but the Pohozaev identity is also valid for multiple ones.

Proof of Corollary 1.5. (a) The Hopf-type lemma [18, Theorem 1.4] requires a supersolution, so we pass to a similar domain $s\Omega$ with $s \le 1$. By the equality $\Lambda_1(s\Omega) = \Lambda_1(\Omega) - 2\ln s$ there exists $s \in (0,1]$ for which $\Lambda_1(s\Omega) \ge 0$. Then $\psi(x) = s^{-N/2}\varphi_1(x/s)$ is the principal eigenfunction in $s\Omega$. It is positive [6, Theorem 1.4(iii)], bounded by Lemma 3.2, continuous in $\mathbb{R}^N$ by Theorem 1.11 of the same work and Theorem 1.1, lies in $H(s\Omega)$, and satisfies $E_{\mathrm{L}}(\psi, v) = \Lambda_1(s\Omega)\langle\psi, v\rangle \ge 0$ for all nonnegative $v \in C_c^\infty(s\Omega)$. Since the interior ball condition holds at every boundary point of $s\Omega$, the same Hopf-type lemma gives $\liminf_{\tau\to 0+} \psi(sx_0 - \tau\nu)\ell(\tau)^{-1/2} > 0$ for all $x_0 \in \partial\Omega$. By Lemma 5.5 this limit equals $\beta(\psi)(sx_0) = s^{-N/2}\beta(\varphi_1)(x_0)$. Hence $\beta(\varphi_1) > 0$ on $\partial\Omega$, and by continuity $\min \beta(\varphi_1) > 0$.

(b) The eigenvalue Λ_1 is simple [6, Theorem 1.4(iii)], and by Theorem 1.4

$$\Lambda_1'(0) = -2 \int \beta_1^2 (X\cdot\nu)\,\mathrm{d}\sigma < 0,$$

since $\beta_1^2 \ge \min \beta_1^2 > 0$, while the continuous function $X\cdot\nu \ge 0$ does not vanish identically.

(c) Since $J_t = 1 + t\,\mathrm{div}\, X + O(t^2)$, by the Gauss–Ostrogradsky formula $\mathrm{d}|\Omega_t|/\,\mathrm{d}t|_{t=0} = \int_\Omega \mathrm{div}\, X\,\mathrm{d}x = \int_{\partial\Omega} X\cdot\nu\,\mathrm{d}\sigma$. Functions of the form $X\cdot\nu$ with $X \in C_c^\infty$ are dense in $C(\partial\Omega)$: if $g \in C(\partial\Omega)$, then uniform approximation of the continuous extension of $g\nu$ by smooth fields X gives $X\cdot\nu \to g$ uniformly. The field X_0 equal to x in a neighborhood of $\overline{\Omega}$ and cut off outside it satisfies $\int X_0\cdot\nu = N|\Omega| \ne 0$, so adding a suitable multiple of this field gives density also among the functions with zero mean. Hence $\int \beta_1^2 g\,\mathrm{d}\sigma = 0$ for all g with $\int g\,\mathrm{d}\sigma = 0$, that is, β_1^2 is constant, and since $\beta_1 > 0$ by (a), β_1 is constant as well. For a ball, β_1 is constant by Lemma 7.3, and $\Lambda_1'(0) = -2\beta_1^2 \int X\cdot\nu\,\mathrm{d}\sigma = 0$ for $\int X\cdot\nu\,\mathrm{d}\sigma = 0$. Moreover, $\beta_1^2 = 1/(N|B_R|)$ by Corollary 1.3. □

The ball is a critical point of Λ_1 at fixed volume, which agrees with the Faber–Krahn inequality for L_Δ [6, Corollary 1.6]: among domains of the same volume, Λ_1 is minimal only for the ball [5, Theorem 1.4].

8.2. **Torsional rigidity and the Green function.** The same scheme applies to solutions of the equation with a fixed right-hand side. Let $f \in L^\infty(\mathbb{R}^N)$, and assume that 0 is not an eigenvalue of L_Δ in Ω. For k_0 with $\Lambda_{k_0}(\Omega) > 0$ we have $\Lambda_k(\Omega_t) \ge \Lambda_{k_0}(\Omega_t)$ for $k \ge k_0$, while $|\Lambda_k(\Omega_t) - \Lambda_k(\Omega)| \le C_k|t|$ for every k (see the proof of Lemma 8.2). The numbers $\Lambda_k(\Omega)$ with $k < k_0$ are nonzero, so for small $|t|$ the distance from zero to the spectrum of L_Δ in Ω_t is bounded below. Since the spectrum is discrete and the eigenfunctions form a basis [6, Theorem 1.4], the problem $L_\Delta u_t = f$ in Ω_t, $u_t = 0$ outside Ω_t, has a unique solution $u_t \in H(\Omega_t)$, and $\|u_t\| \le C$. Put $\mathcal{W}(t) = \int_{\Omega_t} f u_t \,\mathrm{d}x$. For $f = 1$ the solution u_t is called the torsion function, and $\mathcal{W}(t)$ the torsional rigidity of Ω_t.

Theorem 8.3 (Hadamard formula for the torsional rigidity)**.** *Let $f \in L^\infty(\mathbb{R}^N)$, let 0 not be an eigenvalue of L_Δ in Ω, and for small $|t|$ let $u_t \in H(\Omega_t)$ be the solution of the problem $L_\Delta u_t = f$ in Ω_t, $u_t = 0$ outside Ω_t. Then the function $\mathcal{W}(t) = \int_{\Omega_t} f u_t \,\mathrm{d}x$ is differentiable at zero and*

$$\mathcal{W}'(0) = 2\int_{\partial\Omega} \beta(u_0)^2 (X \cdot \nu)\,\mathrm{d}\sigma. \tag{8.2}$$

Proof. We repeat the proof of Theorem 1.4, replacing the eigenfunctions by the solutions $u = u_0$ and u_t. By Lemma 3.2, $\|u_t\|_\infty \le C$, while the boundedness of $E_{\mathrm{L}}(u_t, u_t) = \langle f, u_t\rangle$ together with $\|u_t\| \le C$ and Lemma 3.1 implies that the functions $u_t \circ \Phi_t$ are bounded in $H(\Omega)$. The proof of Lemma 8.2 goes through with one change: the limit ψ of the functions $u_t \circ \Phi_t$ along a subsequence satisfies $E_{\mathrm{L}}(\psi, v) = \langle f, v\rangle$ for all $v \in H(\Omega)$, since $\langle f, S_t v\rangle = \int (f \circ \Phi_t) v J_t \to \langle f, v\rangle$ (approximation of f by continuous functions in $L^2(B)$ for a ball B containing Ω_t for small $|t|$), and $\psi = u$ by the uniqueness of the solution. In the same way as in the lemma we obtain $u_t \circ \Phi_t \to u$ in $L^2(\Omega)$ and the uniform convergence $\beta(u_t) \circ \Phi_t \to \beta(u)$ on $\partial\Omega$. We test the equation for u with $u_t \mathbb{1}_A$, and the equation for u_t with $u \mathbb{1}_A$, and subtract. As in the derivation of (8.1),

$$\int_A f(u_t - u)\,\mathrm{d}x = c_N \int_{\Sigma_+}\int_A \frac{u_t(y)u(x)}{|x-y|^N}\,\mathrm{d}x\,\mathrm{d}y - c_N \int_{\Sigma_-}\int_A \frac{u(y)u_t(x)}{|x-y|^N}\,\mathrm{d}x\,\mathrm{d}y,$$

and for $t > 0$, by Lemma 7.1, the right-hand side equals $2t\int_{\partial\Omega} \beta(u)^2 (X\cdot\nu)\,\mathrm{d}\sigma + o(t)$. Finally,

$$\mathcal{W}(t) - \mathcal{W}(0) = \int_A f(u_t - u)\,\mathrm{d}x + \int_{\Sigma_+} f u_t \,\mathrm{d}x - \int_{\Sigma_-} f u \,\mathrm{d}x,$$

and the last two integrals are $O(t\ell(Ct)^{1/2}) = o(t)$ by Lemma 5.6, while the case $t < 0$ reduces to $-X$. □

For the dilation, $f = 1$ and $u = u_0$, formula (8.2) together with (7.1) yields $\mathcal{W}'(0) = 2\|u\|^2 + N\int u$, which coincides with the derivative found from $L_\Delta(u(\cdot/s)) = (L_\Delta u)(\cdot/s) - 2\ln s \cdot u(\cdot/s)$ for $s > 0$. If $\Lambda_1(\Omega) > 0$, then u minimizes the energy functional $E_{\mathrm{L}}(v,v)/2 - \langle f, v\rangle$ on $H(\Omega)$ with minimum value $-\mathcal{W}(0)/2$, and the derivative of this value under the deformation equals $-\int \beta(u)^2 (X\cdot\nu)\,\mathrm{d}\sigma$.

Corollary 8.4 (torsion function)**.** *Let $\Lambda_1(\Omega) > 0$, $f = 1$ and let $u = u_0$ be the torsion function. Then*

(a) *the function u is positive in Ω, and $\beta(u) > 0$ everywhere on $\partial\Omega$;*
(b) *if $X \cdot \nu \ge 0$ on $\partial\Omega$ and $X \cdot \nu \not\equiv 0$, then $\mathcal{W}'(0) > 0$;*
(c) *if $\mathcal{W}'(0) = 0$ for all X with $\int_{\partial\Omega} X\cdot\nu\,\mathrm{d}\sigma = 0$, then $\beta(u)$ is constant on $\partial\Omega$, and for a ball the condition is satisfied and*

$$\beta(u)^2 = \frac{1}{N|B_R|}\Big(\|u\|^2 + \frac{N}{2}\int_{B_R} u\,\mathrm{d}x\Big).$$

Proof. By the maximum principle [6, Theorem 4.8], which holds for $\Lambda_1(\Omega) > 0$, $u \ge 0$. Moreover, u is not identically zero and is continuous on $\mathbb{R}^N$ by Theorem 1.11 of the same work and Theorem 1.1, and the strong maximum principle [29, Corollary 3.12] gives $u > 0$ in Ω. Since $E_{\mathrm{L}}(u, \varphi) = \int \varphi \ge 0$ for every nonnegative test function φ, the Hopf-type lemma [18, Theorem 1.4] gives $\liminf_{\tau\to 0+} u(x_0 - \tau\nu)\ell(\tau)^{-1/2} > 0$ at every point $x_0 \in \partial\Omega$, and by Lemma 5.5 this limit equals $\beta(u)(x_0)$. Statements (b) and (c) follow from (8.2) in the same way as in Corollary 1.5. The torsion function of the ball is radial: the solution is unique, and the problem is unchanged under rotations. Its boundary coefficient is constant. Since $x\cdot\nu = R$ on ∂B_R and $R|\partial B_R| = N|B_R|$, the first equality in (7.1) with $f = 1$ gives the value $\beta(u)^2$. □

For the torsion function of a small ball, it was proved in [18, Theorem 1.2] that $\liminf_{t\to 0+} u(x_0 - t\nu)\ell(t)^{-1/2}$ is finite and positive. The boundary expansion shows that this limit exists, and Corollary 8.4 expresses it in terms of $\|u\|$ and $\int u$.

Polarization of Theorem 8.3 gives the Hadamard formula for the Green function in weak form.

Corollary 8.5 (Green function)**.** *Let* 0 *not be an eigenvalue of* L_Δ *in* Ω*,* f*,* $g \in L^\infty(\mathbb{R}^N)$*, and let* u_t *and* v_t *be the solutions of the Dirichlet problem in* Ω_t *with right-hand sides* f *and* g *for small* $|t|$*. Then* $t \mapsto \langle g, u_t\rangle$ *is differentiable at zero and*

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle g, u_t\rangle\Big|_{t=0} = 2\int_{\partial\Omega} \beta(u_0)\beta(v_0)(X\cdot\nu)\,\mathrm{d}\sigma.$$

Proof. The weak equations for u_t and v_t, tested with v_t and u_t, give $\langle f, v_t\rangle = E_{\mathrm{L}}(u_t, v_t) = \langle g, u_t\rangle$. This quantity is a symmetric bilinear form in (f, g), equal to $(\mathcal{W}_{f+g}(t) - \mathcal{W}_{f-g}(t))/4$, where $\mathcal{W}_{f\pm g}$ is $\mathcal{W}$ of Theorem 8.3 for the right-hand side $f \pm g$. The solution depends linearly on the right-hand side, the boundary coefficient depends linearly on the solution by the uniqueness in (1.1), and the same theorem for the right-hand sides $f \pm g$ gives the derivative

$$\frac{1}{2}\int_{\partial\Omega} (\beta(u_0+v_0)^2 - \beta(u_0-v_0)^2)(X\cdot\nu)\,\mathrm{d}\sigma = 2\int_{\partial\Omega} \beta(u_0)\beta(v_0)(X\cdot\nu)\,\mathrm{d}\sigma.$$

□

If G_t is the kernel of the operator inverse to L_Δ in Ω_t, then $\langle g, u_t\rangle = \iint G_t(x,y)f(y)g(x)\,\mathrm{d}x\,\mathrm{d}y$, and Corollary 8.5 is the weak form of the formal equality

$$\partial_t G_t(x,y)|_{t=0} = 2\int_{\partial\Omega} \beta(G_0(\cdot,x))\beta(G_0(\cdot,y))(X\cdot\nu)\,\mathrm{d}\sigma.$$

This is an analogue of the Hadamard formula for the Green function of the Laplacian, in which $2\beta(\cdot)\beta(\cdot)$ is replaced by the product of the normal derivatives. The pointwise form requires extending the boundary expansion to solutions with a Dirac measure on the right-hand side, and it is not considered here. For the fractional Laplacian, such a formula was obtained by Djitte and Sueur [12, Theorems 1.2, 1.5].

8.3. **Off-center hole.** For the fractional Laplacian, Djitte, Fall and Weth proved that the principal eigenvalue and the torsional rigidity of a domain $B_1 \setminus \overline{B_\tau(a)}$ are extremal for $a = 0$ [10, Theorem 1.4] (for the torsional rigidity this is the case $p = 1$ of their theorem), in analogy with Hersch's theorem for the principal eigenvalue of the Laplacian [19]. Their argument, based on the Hadamard formula and maximum principles for antisymmetric functions, carries over to L_Δ with Theorems 1.4 and 8.3.

Theorem 8.6 (off-center hole)**.** *Let* $N \ge 2$*,* $0 < \tau < R$ *and* $\Omega^t = B_R \setminus \overline{B_\tau(te_1)}$ *for* $0 \le t < R - \tau$*. Then* $\Lambda_1(\Omega^t)$ *is strictly decreasing in* t*. If* $\Lambda_1(B_R) > 0$*, then the torsional rigidity* $\int u^t\,\mathrm{d}x$ *of* Ω^t*, where* $L_\Delta u^t = 1$ *in* Ω^t *and* $u^t = 0$ *outside* Ω^t*, is strictly increasing in* t*.*

Proof. We express the derivatives in terms of the boundary coefficients on the sphere of the hole and compare them at pairs of points obtained from one another by reflection in the plane $P_t = \{x_1 = t\}$, using the maximum principles for antisymmetric functions [29].

Derivatives. Fix $t \in (0, R-\tau)$ and a function $\eta \in C_c^\infty(B_R)$ equal to one in a neighborhood of $\overline{B_\tau(te_1)}$. For $X = \eta e_1$ with small $|s|$ we have $\Phi_s(\Omega^t) = \Omega^{t+s}$, and on the sphere $S = S_\tau(te_1)$ the outward normal to Ω^t equals $-(x - te_1)/\tau$, and $X\cdot\nu = -(x_1 - t)/\tau$. The reflection Q in the plane P_t maps S to itself, and for a function β continuous on S

$$\int_S \beta^2(x_1 - t)\,\mathrm{d}\sigma = -\int_{S\cap\{x_1<t\}} (t - x_1)(\beta(x)^2 - \beta(Qx)^2)\,\mathrm{d}\sigma.$$

The principal eigenvalue is simple [6, Theorem 1.4(iii)], and by Theorem 1.4 the derivative of $\Lambda_1(\Omega^t)$ with respect to t equals $(2/\tau)\int_S \beta^2(x_1 - t)\,\mathrm{d}\sigma$ with $\beta = \beta(\varphi_1)$, where $\varphi_1 > 0$ is the normalized principal eigenfunction. By Theorem 8.3 the derivative of the torsional rigidity equals $-(2/\tau)\int_S \beta^2(x_1 - t)\,\mathrm{d}\sigma$ with $\beta = \beta(u^t)$. It suffices to prove that in both cases $\beta(x) > \beta(Qx)$ for $x \in S$, $x_1 < t$. Because $\beta > 0$ on $\partial\Omega^t$ (Corollaries 1.5(a) and 8.4(a), where the condition $\Lambda_1(\Omega^t) > 0$ of the latter is verified below), it follows that $\beta(x)^2 > \beta(Qx)^2$, and on $(0, R-\tau)$ the derivative of $\Lambda_1(\Omega^t)$ is negative and the derivative of the torsional rigidity is positive. Since both functions are continuous at $t = 0$ (the argument above at $t = 0$, with $\eta = 1$ near the concentric hole, gives their differentiability), they are strictly monotone on all of $[0, R-\tau)$.

Antisymmetric difference. Let $\mathbb{H} = \{x_1 < t\}$, $\varphi = \varphi_1$ or $\varphi = u^t$, $w = \varphi - \varphi\circ Q$ and $\Omega_* = \mathbb{H}\cap\Omega^t\cap Q(\Omega^t)$. For $x \in \mathbb{H}$ we have $|Qx|^2 = |x|^2 + 4t(t - x_1) > |x|^2$, and Q maps the ball $B_\tau(te_1)$ onto itself. Therefore the set $\Omega_* = (\mathbb{H}\cap Q(B_R)) \setminus \overline{B_\tau(te_1)}$ is open and Lipschitz, $w = 0$ in $\mathbb{H}\setminus\Omega^t$ and $w = \varphi > 0$ in $(\mathbb{H}\cap\Omega^t)\setminus Q(\Omega^t)$. Here $\varphi > 0$: for the torsion function this follows from Corollary 8.4(a), since $\Omega^t \subset B_R$, the extension by zero embeds $H(\Omega^t)$ into $H(B_R)$, and therefore $\Lambda_1(\Omega^t) \ge \Lambda_1(B_R) > 0$.

Moreover, w is antisymmetric, $w \circ Q = -w$, and continuous on $\mathbb{R}^N$ [6, Theorem 1.11 and (1.10)], and L_Δ commutes with reflections. Since $Q(\Omega_*) \subset \Omega^t$, the function w is a weak solution in Ω_* of the equation $L_\Delta w = \Lambda w$ with $\Lambda = \Lambda_1(\Omega^t)$ in the first case and $L_\Delta w = 0$, that is, $\Lambda = 0$, in the second. In the first case a dilation shifts Λ_1 by a constant (Section 6) and multiplies the boundary coefficients by a common factor, so one may assume that $\Lambda \geq 0$. Moreover, $\Lambda_1(\Omega_*) > \Lambda_1(\Omega^t)$: otherwise the principal eigenfunction of Ω_*, extended by zero, would minimize the Rayleigh quotient in Ω^t and would coincide with $\pm\varphi_1$, while $\varphi_1 > 0$ in Ω^t. Hence $0 \leq \Lambda < \Lambda_1(\Omega_*)$, which is exactly the condition $\|V\|_\infty < \Lambda_1(\Omega_*)$ of the antisymmetric weak maximum principle [29, Proposition 3.1] with $V = \Lambda$. In the second case $V = 0$, and the condition holds since $\Omega_* \subset B_R$ and $\Lambda_1(\Omega_*) \geq \Lambda_1(B_R) > 0$.

Maximum principles. By the antisymmetric weak maximum principle, $w \geq 0$ in $\mathbb{H}$. Moreover, w is positive on the set $(\mathbb{H} \cap \Omega^t) \setminus Q(\Omega^t)$ of positive measure, and the antisymmetric strong maximum principle [29, Corollary 3.6] gives $w > 0$ in Ω_*. Let $x \in S$ with $x_1 < t$, let $B \subset \Omega_*$ be a ball of small radius tangent to S at x with $\Lambda_1(B) > \Lambda$ (since $\Lambda_1(B_r) = \Lambda_1(B_1) - 2\ln r \to \infty$ as $r \to 0$), and let $B' = B_{\rho'}(y') \subset\subset \Omega_* \setminus \overline{B}$ be a ball on which $w \geq c > 0$ (it exists because $w > 0$ is continuous in Ω_*). For a small radius, $\overline{B} \subset \mathbb{H}$ and $\Lambda_1(B) > 0$, and $w(x) = 0$ since $x \in S$. These are the hypotheses of the antisymmetric Hopf lemma [29, Lemma 3.4], which gives

$$\liminf_{s \to 0+} w(x - s\nu(x))\ell(s)^{-1/2} > 0.$$

The reflection Q is an isometry fixing the point te_1, $\nu(Qx)$ is the image of $\nu(x)$ under the linear part of Q, and $Q(x - s\nu(x)) = Qx - s\nu(Qx)$. By Lemma 5.5 this lower limit equals $\beta(\varphi)(x) - \beta(\varphi)(Qx)$, and $\beta(x) > \beta(Qx) \geq 0$, as required. □

8.4. **Multiple eigenvalues and generic simplicity.** For a multiple eigenvalue two distinct eigenfunctions occur in the thin layers, and a polarized form of Lemma 7.1 is needed.

Lemma 8.7 (polarization)**.** *Let w, $\tilde{w} \in H(\Omega)$ be weak solutions of the problems (2.2) with bounded right-hand sides, and let w_h satisfy the assumptions of Lemma 7.1(b) with $\beta(w_h) \circ \Phi_h \to \beta(w)$. Then as $h \to 0+$*

$$\begin{aligned} c_N \int_{\Sigma_+} w_h(y) \int_A \frac{\tilde{w}(x)}{|x-y|^N} \,\mathrm{d}x \,\mathrm{d}y &= 2h \int_{\partial\Omega} \beta(w)\beta(\tilde{w})(X \cdot \nu)_+ \,\mathrm{d}\sigma + o(h), \\ c_N \int_{\Sigma_-} \tilde{w}(y) \int_A \frac{w_h(x)}{|x-y|^N} \,\mathrm{d}x \,\mathrm{d}y &= 2h \int_{\partial\Omega} \beta(w)\beta(\tilde{w})(X \cdot \nu)_- \,\mathrm{d}\sigma + o(h). \end{aligned} \tag{8.3}$$

Proof. By Lemma 7.1(b), applied to $\tilde{w}$ in Ω and to w_h in Ω_h, the assumptions of its part (a) hold with $u = \tilde{w}$, $v = w_h$, $\mathcal{J}(x-y) = c_N|x-y|^{-N}$, $\alpha = \beta(\tilde{w})$ and $\gamma = \beta(w)$. Part (a) gives (8.3). □

Theorem 8.8 (Hadamard formula for a multiple eigenvalue)**.** *Let Λ be an eigenvalue of L_Δ in Ω of multiplicity m, let $\varphi_1, \dots, \varphi_m$ be a real orthonormal basis of the eigenspace, $X \in C^{1,1}(\mathbb{R}^N; \mathbb{R}^N)$,*

$$M_{ij} = -2 \int_{\partial\Omega} \beta(\varphi_i)\beta(\varphi_j)(X \cdot \nu) \,\mathrm{d}\sigma,$$

and let $\eta_1 \leq \dots \leq \eta_m$ be the eigenvalues of the symmetric matrix M. Then for small $|t|$ the operator L_Δ in Ω_t has exactly m eigenvalues near Λ counted with multiplicity, $\Lambda^{(1)}(t) \leq \dots \leq \Lambda^{(m)}(t)$, and

$$\lim_{t \to 0+} \frac{\Lambda^{(k)}(t) - \Lambda}{t} = \eta_k, \quad \lim_{t \to 0-} \frac{\Lambda^{(k)}(t) - \Lambda}{t} = \eta_{m+1-k}, \quad k = 1, \dots, m.$$

Proof. We estimate the rate of convergence of $\Lambda^{(k)}(t)$, pass to the limit in the eigenfunctions, and from the cross identity find the one-sided derivatives of $\Lambda^{(k)}$ at zero as the eigenvalues of the matrix M. If $\Lambda = \Lambda_{j+1}(\Omega) = \dots = \Lambda_{j+m}(\Omega)$ and $\Lambda_j(\Omega) < \Lambda < \Lambda_{j+m+1}(\Omega)$ (for $j = 0$ we set $\Lambda_0 = -\infty$), then, as in the proof of Lemma 8.2, $|\Lambda_i(\Omega_t) - \Lambda_i(\Omega)| \leq C_i|t|$ for each i, so that $\Lambda^{(k)}(t) = \Lambda_{j+k}(\Omega_t)$ and $|\Lambda^{(k)}(t) - \Lambda| \leq C|t|$. Let $t > 0$, and let $\psi_1^t, \dots, \psi_m^t$ be real orthonormal eigenfunctions in Ω_t with eigenvalues $\Lambda^{(k)}(t)$. By Lemma 3.2, in which the constant is nondecreasing in μ, $\|\psi_k^t\|_\infty \leq C$.

Limits. As in the proof of Lemma 8.2, every sequence $t_n \to 0+$ contains a subsequence along which $\psi_k^t \circ \Phi_t \to \psi_k$ in $L^2(\Omega)$, $(\Lambda^{(k)}(t) - \Lambda)/t \to \tilde{\eta}_k$ and $\beta(\psi_k^t) \circ \Phi_t \to \beta(\psi_k)$ uniformly on $\partial\Omega$ (the argument of the lemma is applied to the finite family after choosing a diagonal subsequence), and the limits ψ_k are eigenfunctions with eigenvalue Λ. Since $\langle S_t u, S_t v\rangle - \langle u, v\rangle = O(t)\|u\|\|v\|$ (Lemma 8.1), $\psi_1, \dots, \psi_m$ are orthonormal and form a basis of the eigenspace, that is, $\psi_k = \sum_i o_{ki}\varphi_i$ with an orthogonal matrix $O = (o_{ki})$, and $\tilde{\eta}_1 \leq \dots \leq \tilde{\eta}_m$.

Cross identity. As in the derivation of (8.1), for all k and i

$$(\Lambda^{(k)}(t)-\Lambda)\langle\psi_k^t,\varphi_i\mathbb{1}_A\rangle=-c_N\int_{\Sigma_+}\int_A\frac{\psi_k^t(y)\varphi_i(x)}{|x-y|^N}\,\mathrm{d}x\,\mathrm{d}y+c_N\int_{\Sigma_-}\int_A\frac{\varphi_i(y)\psi_k^t(x)}{|x-y|^N}\,\mathrm{d}x\,\mathrm{d}y.$$

By (8.3) with $w=\psi_k$, $w_h=\psi_k^t$ and $\tilde w=\varphi_i$, the right-hand side equals $-2t\int\beta(\psi_k)\beta(\varphi_i)(X\cdot\nu)\,\mathrm{d}\sigma+o(t)$, and $\langle\psi_k^t,\varphi_i\mathbb{1}_A\rangle\to o_{ki}$. Since the boundary coefficient is linear on the eigenspace, the vector $(\beta(\psi_k))_k$ equals $O(\beta(\varphi_i))_i$, and after dividing by t we obtain

$$\tilde\eta_k o_{ki}=(OM)_{ki},$$

that is, $OMO^\top=\operatorname{diag}(\tilde\eta_1,\dots,\tilde\eta_m)$. Hence, $\tilde\eta_1\le\dots\le\tilde\eta_m$ are the eigenvalues of M in increasing order, $\tilde\eta_k=\eta_k$, and the limit does not depend on the subsequence. For $t<0$ we have $\Phi_t=\mathrm{id}+|t|(-X)$, and the field $-X$ gives the matrix $-M$ with eigenvalues $-\eta_m\le\dots\le-\eta_1$, and $(\Lambda^{(k)}(t)-\Lambda)/|t|\to-\eta_{m+1-k}$, that is, $(\Lambda^{(k)}(t)-\Lambda)/t\to\eta_{m+1-k}$. □

For $m=1$, Theorem 8.8 coincides with Theorem 1.4.

For linear fields $X(x)=Bx$, the matrix M is expressed in terms of the solution in the domain. For the fractional Laplacian the Pohozaev identity with an arbitrary Lipschitz field was proved in [11, Theorem 1.1], and the following statement is an analogue of the particular case $X=Bx$ of that identity.

Proposition 8.9 (matrix Pohozaev identity)**.** *Let z be a weak solution of the problem $L_\Delta z=\mu z+f$ in Ω, $z=0$ outside Ω, with $f\in L^\infty(\Omega)\cap W^{1,1}(\Omega)$, and $B\in\mathbb{R}^{N\times N}$. Then*

$$\int_{\partial\Omega}\beta(z)^2(Bx\cdot\nu)\,\mathrm{d}\sigma=\langle P_Bz,z\rangle+\frac{\operatorname{tr}B}{2}\langle f,z\rangle+\langle z,Bx\cdot\nabla f\rangle,\tag{8.4}$$

where P_B is the operator with symbol $(Bk\cdot k)/|k|^2$.

Proof. Let us repeat the derivation of the Pohozaev identity, replacing the dilation by a linear map. The form then changes by a Fourier multiplier, and the thin layers require part (a) of Lemma 7.1, whose assumptions for the solution in the deformed domain are verified separately.

Let $h>0$ be small, $\Phi_h(x)=(I+hB)x$, $J_h=\det(I+hB)$, $\Omega_h=\Phi_h(\Omega)$ and $T_hw=J_h^{-1/2}w\circ\Phi_h^{-1}$. The operator T_h unitarily maps $L^2(\Omega)$ onto $L^2(\Omega_h)$, and its action on the Fourier transform is $\widehat{T_hw}(k)=J_h^{1/2}\hat w((I+hB)^\top k)$. The substitution $k'=(I+hB)^\top k$ in the symbol representation of the form gives

$$E_{\mathrm L}(T_hu,T_hv)=E_{\mathrm L}(u,v)+\langle Q_hT_hu,T_hv\rangle,\quad\widehat{Q_hw}(k)=q_h(k)\hat w(k),\quad q_h(k)=-\ln\frac{|(I+hB)^\top k|^2}{|k|^2},$$

with $|q_h(k)+2h(Bk\cdot k)/|k|^2|\le Ch^2$ uniformly in k. For $B=I$ this equality coincides with (6.1). As in the proof of Lemma 6.1, $z_2=T_hz\in H(\Omega_h)$ satisfies the equation $L_\Delta z_2=\mu z_2+T_hf+Q_hz_2$ in Ω_h with bounded test functions, in particular with $z\mathbb{1}_A$, and subtracting the equations, as in Lemma 6.2, gives

$$-c_N\int_{\Sigma_+}\int_A\frac{z_2(y)z(x)}{|x-y|^N}\,\mathrm{d}x\,\mathrm{d}y+c_N\int_{\Sigma_-}\int_A\frac{z(y)z_2(x)}{|x-y|^N}\,\mathrm{d}x\,\mathrm{d}y=\langle T_hf,z\mathbb{1}_A\rangle-\langle f,z_2\mathbb{1}_A\rangle+\langle Q_hz_2,z\mathbb{1}_A\rangle.$$

Left-hand side. Part (b) of Lemma 7.1 does not apply to z_2: Theorem 1.1 in Ω_h requires a bounded right-hand side, and the boundedness of Q_hz_2 is unknown. We apply part (a) with $u=z$, $v=z_2$, $\mathcal J(x-y)=c_N|x-y|^{-N}$, $\alpha=\gamma=\beta=\beta(z)$ and $X(x)=Bx$. The conditions on u involve only z and hold by part (b). Let $K=\|z\|_\infty+\|\mu z+f\|_\infty$, $d_h(y)=\operatorname{dist}(y,\mathbb{R}^N\setminus\Omega_h)$ and $\delta_h(y)=\operatorname{dist}(y,\Omega_h)$, and in the chart let $y_0=(y',\phi(y'))$ and $\eta=\phi_h-\phi$, as in the proof of the lemma. The conditions on v for $y^*=y_0$ (here $|y-y_0|\le|\eta(y')|\le Ch$) are the estimate $\|z_2\|_\infty\le CK$, the equality

$$z_2(y)=\beta(y_0)\zeta^{-1/2}+O((\varepsilon_h'+K\zeta^{-1/2})\zeta^{-1/2}),\qquad\zeta=\ln\frac{1}{d_h(y)},\tag{8.5}$$

for almost all $y\in\Sigma_+$, and the formula for the outer potential

$$\mathcal P_h(y)=c_N\int_{\Omega_h}\frac{z_2(x)}{|x-y|^N}\,\mathrm{d}x=2\beta(y_0)\Big(\ln\frac{1}{\delta_h}\Big)^{1/2}+O\Big(K+\varepsilon_h'\Big(\ln\frac{1}{\delta_h}\Big)^{1/2}+Kh\ln\frac{1}{\delta_h}\Big)\tag{8.6}$$

on Σ_- outside a set of measure zero, where $\varepsilon_h'\to0$ as $h\to0$.

Let us verify them, assuming that C also depends on the operator norm $\|B\|$, and that h is so small that $b_h=h\|B\|\le1/2$. Then the norms of Φ_h and Φ_h^{-1} are bounded by $1+2b_h$, and together with the

fact that Φ_h maps Ω and $\mathbb{R}^N\setminus\Omega$ to Ω_h and $\mathbb{R}^N\setminus\Omega_h$, this gives

$$\frac{d(x)}{1+2b_h}\le d_h(\Phi_h(x))\le(1+2b_h)d(x),\qquad \frac{\operatorname{dist}(x,\Omega)}{1+2b_h}\le\delta_h(\Phi_h(x))\le(1+2b_h)\operatorname{dist}(x,\Omega),$$

and the difference between the logarithms of the corresponding distances does not exceed $2b_h$ in absolute value. For $y\in\Sigma_\pm$ we have $|\Phi_h^{-1}(y)-y|\le Ch$ (here C also depends on $\sup_\Omega|x|$) and $|y-y_0|\le|\eta(y')|\le Ch$, and the estimate $\|z_2\|_\infty\le CK$ follows from $|z_2|\le J_h^{-1/2}\|z\|_\infty\le 2^{N/2}K$.

Values in the layer. Let $y\in\Sigma_+$ and $\tilde x=\Phi_h^{-1}(y)\in\Omega$. Then $d(\tilde x)\le(1+2b_h)d_h(y)\le Ch<r_*$, and by (1.1) for almost all y

$$z(\tilde x)=\beta(\pi(\tilde x))\xi^{-1/2}+O(K\xi^{-1}),\qquad \xi=\ln\frac{1}{d(\tilde x)},\quad |\xi-\zeta|\le 2b_h.$$

Furthermore, $|\pi(\tilde x)-y_0|\le d(\tilde x)+|\tilde x-y|+|y-y_0|\le Ch$, and by (1.2) $|\beta(\pi(\tilde x))-\beta(y_0)|\le\varpi(Ch)$. Taking into account the identities $z_2(y)=J_h^{-1/2}z(\tilde x)$, $J_h^{-1/2}=1+O(h)$ and the bounds $|\beta|\le CK$, $|\xi^{-1/2}-\zeta^{-1/2}|\le Ch\zeta^{-3/2}$, we obtain (8.5) with $\varepsilon_h'=\varpi(Ch)+CKh$.

Outer potential. Let $y\in\Sigma_-\setminus\partial\Omega_h$, $\tilde y=\Phi_h^{-1}(y)\notin\overline\Omega$ and $\tilde\delta=\operatorname{dist}(\tilde y,\Omega)\le(1+2b_h)\delta_h(y)\le Ch$. The substitution $x=\Phi_h(\tilde x)$ gives

$$\mathcal{P}_h(y)=J_h^{1/2}c_N\int_\Omega\frac{z(\tilde x)}{|\Phi_h(\tilde x-\tilde y)|^N}\,\mathrm{d}\tilde x,$$

while $|\Phi_h(v)|^{-N}=|v|^{-N}(1+O(h))$ uniformly in v. The bound $c_N\int_\Omega|\tilde x-\tilde y|^{-N}\,\mathrm{d}\tilde x\le 2\ln(2D/\tilde\delta)$ implies $\mathcal{P}_h(y)=J_h^{1/2}\mathcal{P}(\tilde y)+O(Kh(1+\ln(1/\tilde\delta)))$, where $\mathcal{P}(\tilde y)=c_N\int_\Omega z(\tilde x)|\tilde x-\tilde y|^{-N}\,\mathrm{d}\tilde x$. By Lemma 5.1(d) and (5.5), applicable for small $\tilde\delta$,

$$\mathcal{P}(\tilde y)=U_p\Big(\ln\frac{1}{\tilde\delta}\Big)+O(K)=2\beta(p)\Big(\ln\frac{1}{\tilde\delta}\Big)^{1/2}+O(K),$$

where $p\in\partial\Omega$ is the point closest to $\tilde y$, $|p-y_0|\le\tilde\delta+|\tilde y-y|+|y-y_0|\le Ch$. It remains to replace $\beta(p)$ by $\beta(y_0)$, $\ln(1/\tilde\delta)$ by $\ln(1/\delta_h)$ and $J_h^{1/2}$ by one. The errors of these replacements are $\varpi(Ch)$, at most $2b_h$ and $O(h)$, which gives (8.6).

By Lemma 7.1(a), the left-hand side equals $-2h\int\beta^2(Bx\cdot\nu)\,\mathrm{d}\sigma+o(h)$.

Right-hand side. The proof of Lemma 7.2 with this Φ_h goes through without changes: for $d(y)>Ch$ the curves $s\mapsto(I+sB)^{\pm1}y$, $0\le s\le h$, lie in Ω, and the difference quotients of f along them converge in $L^1(\{d>\delta\})$ for each $\delta>0$, the derivative of $J_h^{\mp1/2}$ at $h=0$ equals $\mp\operatorname{tr}B/2$, and the derivatives of $f(\Phi_h^{\mp1}x)$ equal $\mp Bx\cdot\nabla f$. Therefore, the first two terms, divided by h, tend to $-\operatorname{tr}B\langle f,z\rangle-2\langle z,Bx\cdot\nabla f\rangle$. Since $\|Q_h/h+2P_B\|_{L^2\to L^2}\le Ch$, $z_2\to z$ and $z\mathbb{1}_A\to z$ in L^2, the last term, divided by h, tends to $-2\langle P_Bz,z\rangle$. Dividing by $-2h$ and passing to the limit gives (8.4). □

For $B=I$ we have $P_I=I$, and (8.4) is (1.3) for $a=0$. For eigenfunctions φ_i, polarizing (8.4) with $f=0$ gives $M_{ij}=-2\langle P_B\varphi_i,\varphi_j\rangle$ for $X(x)=Bx$, so that for linear fields the matrix of Theorem 8.8 is expressed without boundary coefficients.

Together with Corollary 1.3, Theorem 8.8 shows that a multiple eigenvalue splits under a suitable small deformation of the boundary.

Lemma 8.10 (splitting)**.** *If Λ is an eigenvalue of L_Δ in Ω of multiplicity $m\ge2$, then there exists a field $Y\in C_c^\infty(\mathbb{R}^N;\mathbb{R}^N)$ for which the matrix M of Theorem 8.8 is not proportional to the identity.*

Proof. If $M=c(Y)I$ for all Y, then $\int\beta(\varphi_1)\beta(\varphi_2)(Y\cdot\nu)\,\mathrm{d}\sigma=0$ and $\int(\beta(\varphi_1)^2-\beta(\varphi_2)^2)(Y\cdot\nu)\,\mathrm{d}\sigma=0$ for all Y. Functions of the form $Y\cdot\nu$ are dense in $C(\partial\Omega)$ (the proof of Corollary 1.5(c)), and the boundary coefficients are continuous, so $\beta(\varphi_1)\beta(\varphi_2)\equiv0$ and $\beta(\varphi_1)^2\equiv\beta(\varphi_2)^2$ on $\partial\Omega$. Hence $\beta(\varphi_1)^4=\beta(\varphi_1)^2\beta(\varphi_2)^2\equiv0$, which contradicts Corollary 1.3. □

Theorem 8.11 (generic simplicity of the spectrum)**.** *Let $\mathcal{X}$ be the open ball of radius ε_0 centered at zero in the Banach space $C^{1,1}(\mathbb{R}^N;\mathbb{R}^N)$ with norm $\sup|X|+\sup|\nabla X|+\operatorname{Lip}(\nabla X)$, where $0<\varepsilon_0\le1/20$. Those $X\in\mathcal{X}$ for which all eigenvalues of L_Δ in $(\mathrm{id}+X)(\Omega)$ are simple form a dense G_δ subset of $\mathcal{X}$. The same holds for $(I-\Delta)^{\log}$.*

Proof. For $X\in\mathcal{X}$ the mapping $\mathrm{id}+X$ is a diffeomorphism of $\mathbb{R}^N$ of class $C^{1,1}$: the equation $x+X(x)=y$ is solved by the contraction $x\mapsto y-X(x)$ with constant $1/20$, and the smoothness of the inverse mapping is given by the inverse function theorem. The domain $(\mathrm{id}+X)(\Omega)$ satisfies the two-sided ball condition with radius bounded below uniformly in $X\in\mathcal{X}$: in charts its boundary is given by graphs of functions whose

$C^{1,1}$ norms are bounded through the norms of the charts Ω, since $\|\nabla X\|_\infty \le 1/20$ and $\mathrm{Lip}(\nabla X) \le 1/20$ (cf. Section 7.1). We prove that for each n the set $\mathcal{X}_n$ of fields for which the first n eigenvalues are simple is open and dense, and apply the Baire theorem.

Openness. Set $\Omega_X = (\mathrm{id}+X)(\Omega)$ and $\mathcal{X}_n = \{X \in \mathcal{X} : \Lambda_k(\Omega_X) < \Lambda_{k+1}(\Omega_X),\ k = 1, \dots, n\}$. The proof of Lemma 8.1 uses only the smallness of $\|\nabla(\Phi - \mathrm{id})\|_\infty$ and therefore goes through for the diffeomorphism $\Phi = (\mathrm{id}+X') \circ (\mathrm{id}+X)^{-1}$, where $X' \in \mathcal{X}$ and $\|X' - X\|_{C^1}$ is small. The minimax principle gives

$$|\Lambda_k(\Omega_{X'}) - \Lambda_k(\Omega_X)| \le C_k \|X' - X\|_{C^1},$$

so that the set $\mathcal{X}_n$ is open.

Density. Fix $X_0 \in \mathcal{X}$ and an open $U \ni X_0$. Since $\Lambda_k \to \infty$, there exists $n' \ge n$ with $\Lambda_{n'}(\Omega_{X_0}) < \Lambda_{n'+1}(\Omega_{X_0})$. Denote by $p(X)$ the number of indices $k \le n'$ with $\Lambda_k(\Omega_X) = \Lambda_{k+1}(\Omega_X)$.

If $p(X_0) > 0$, we take an eigenvalue Λ of the operator L_Δ in Ω_{X_0} of multiplicity $m \ge 2$ with indices not exceeding n', and let Y be the field of Lemma 8.10 for Ω_{X_0}. By Theorem 8.8, for small $t > 0$ the eigenvalues of $(\mathrm{id}+tY)(\Omega_{X_0})$ near Λ have right-hand derivatives $\eta_1 \le \dots \le \eta_m$, and $\eta_1 < \eta_m$, so that these eigenvalues split into groups of smaller multiplicity, while for sufficiently small t the gaps between the remaining distinct groups among the first $n'+1$ eigenvalues do not close. Within the group of Λ at most $m-2$ equalities remain, since $\eta_1 < \eta_m$. In every other group the number of equalities does not exceed its multiplicity minus one, and different groups stay separated. Here $(\mathrm{id}+tY)(\Omega_{X_0})$ equals Ω_{X_1} with $X_1 = X_0 + tY \circ (\mathrm{id}+X_0)$, and $X_1 \in U$ for small t, while $p(X_1) < p(X_0)$. After finitely many steps we obtain $X \in U$ with $p(X) = 0$, that is, $X \in \mathcal{X}_n$. Hence $\mathcal{X}_n$ is dense, and by the Baire category theorem the intersection $\bigcap_n \mathcal{X}_n$ is dense and of type G_δ. □

For $(I-\Delta)^{\log}$ the proof of Theorem 8.11 is completed in Section 9.3, and for the Dirichlet Laplacian the generic simplicity under deformation of the domain was proved by Micheletti [25, 26] and Uhlenbeck [33, Theorem 9]. In the fractional case, Fall, Ghimenti, Micheletti and Pistoia [13, Theorem 1] proved that by an arbitrarily small deformation of the domain all eigenvalues can be made simple. A localized deformation of a Lipschitz domain splitting the spectrum of the Laplacian was constructed by Dabrowski [9].

9. Operators of logarithmic order

For $b \equiv c_N$ the proofs of the results carried over here use the exact form of the kernel $c_N|z|^{-N}$ only in two places, while for anisotropic b the bump functions, the spherical means, the layers and the transport of the form change as well. In the truncation, the mass of the kernel in the layer $r \le |z| < 1$ gives the factor $2\ln(1/r)$, and in the dilation, the homogeneity of the kernel of degree $-N$ is used. The remaining terms of the form enter the equation as a bounded right-hand side. Therefore the results listed in Theorem 1.6 carry over to operators with form (1.5), and its proof consists of the following changes. Let b, V, κ satisfy the conditions (B1)–(B3) of Section 1 and $\theta = (x-y)/|x-y|$.

9.1. **Boundedness and boundary expansion.**

Proof of Theorem 1.6(a). Since $0 < \min b \le b \le \max b$, the local part of the form (1.5) is equivalent to $\mathcal{E}$, and the space $H(\Omega)$ is unchanged. The forms $\langle Vu, v\rangle$ and $\langle \kappa * u, v\rangle$ are bounded in $L^2(\Omega)$, since $\kappa \mathbb{1}_{B_D} \in L^1$, while the values of κ outside B_D do not enter the problem.

Truncation. Lemma 3.1 holds with $c_N|z|^{-N}$ replaced by $b(\theta)|z|^{-N}$, the term $\rho_N\langle u, v\rangle$ by $\langle Vu, v\rangle$ and with the addition of $\langle \kappa * u, v\rangle$, since for $r < 1$ the mass of the kernel in the layer $r \le |z| < 1$ equals $(\int b\,\mathrm{d}\omega)\ln(1/r) = 2\ln(1/r)$, and for $r > 1$ the computation for $1 \le |z| < r$ repeats its proof.

Boundedness. Let $Lu = \mu u + g$ weakly, $g \in L^\infty$. Choose M so that $\kappa_1 = \kappa \mathbb{1}_{B_D} \mathbb{1}_{|\kappa|>M}$ has $\|\kappa_1\|_{L^1} \le 1/4$, and set $\kappa_2 = \kappa \mathbb{1}_{B_D} - \kappa_1$, where $|\kappa_2| \le M$. For $t \ge 0$ set $v = (u-t)_+$, $\tilde v = (-u-t)_+$, $Z = \|v\|^2 + \|\tilde v\|^2$, $I = \int (v+\tilde v)$, $m_r = 2\ln(1/r) - |\mu| - \|V\|_\infty$ and $j_{b,r}(z) = b(z/|z|)|z|^{-N}\mathbb{1}_{|z|\ge r}$. As in the proof of Lemma 3.2, the equations with test functions v and $-\tilde v$ give

$$m_r(Z + tI) \le (\|j_{b,r}\| + \|\kappa_2\|_{L^2})\|u\|I + \|g\|_\infty I + \langle |\kappa_1| * |u|, v + \tilde v\rangle.$$

Since $|u| \le t + v + \tilde v$, the last term does not exceed $\|\kappa_1\|_{L^1}(tI + 2Z) \le (tI + 2Z)/4$. Choose $r = \exp(-(1+|\mu|+\|V\|_\infty)/2)$ so that $m_r = 1$, and set $t = 2[(\|j_{b,r}\|+\|\kappa_2\|_{L^2})\|u\|+\|g\|_\infty]$. Then $Z/2 + tI/4 \le 0$, that is, $Z = 0$, and $\|u\|_\infty \le C(\|u\| + \|g\|_\infty)$, where C depends on N, D, μ, b, $\|V\|_\infty$ and κ. As $|\mu|$ increases, the number r decreases, the norm $\|j_{b,r}\|$ grows, and κ_2 is chosen independently of μ, therefore C is nondecreasing in $|\mu|$.

Reduction. If u is bounded, then $L_b u = g + \mu u - Vu - \kappa * u$ with a bounded right-hand side, where L_b is the operator of form (1.5) with $V = 0$ and $\kappa = 0$, and it suffices to prove Theorem 1.1 for L_b.

Value and potential. In part (a) of Lemma 4.1 the potential is replaced by

$$\mathcal{V}_b(x)=\int_{|z-x|\ge\delta/2} u(z)b((z-x)/|z-x|)|z-x|^{-N}\,\mathrm{d}z,$$

and the bump function by $\varphi_s(y)=d_s^{-N}\chi_0(|y-x|/d_s)|S^{N-1}|b((y-x)/|y-x|)/2$. Then $\alpha(s)$ is expressed in terms of the weighted averages $\bar u_b(\rho)=(1/2)\int_{S^{N-1}} b(\omega)u(x+\rho\omega)\,\mathrm{d}\omega$ with the same weight w, and the far term gives $2\int_0^s \tilde u_b$ instead of $2\int_0^s\tilde u$, where $\tilde u_b(\sigma)=\bar u_b((\delta/4)e^{-\sigma})$. Set $\mathcal{J}_b(v)=b(v/|v|)|v|^{-N}$, so that $\mathcal{V}_b(x)=\int_{|z-x|\ge\delta/2}u(z)\mathcal{J}_b(z-x)\,\mathrm{d}z$. Since $|v/|v|-v'/|v'||\le 2|v-v'|/|v|$ and $||v|^{-N}-|v'|^{-N}|\le N2^{N+1}|v-v'||v|^{-N-1}$ for $|v-v'|\le|v|/2$, it follows that

$$|\mathcal{J}_b(v)-\mathcal{J}_b(v')|\le C_b|v-v'||v|^{-N-1}\quad(|v-v'|\le|v|/2),\qquad C_b=N2^{N+1}\max b+2\,\mathrm{Lip}(b),\tag{9.1}$$

where $\mathrm{Lip}(b)$ is taken with respect to the Euclidean distance on S^{N-1}, and the mass of the kernel $\mathcal{J}_b$ in the layer $a<|v|<a'$ equals $2\ln(a'/a)$, as for $c_N|v|^{-N}$. The function φ_s is nonnegative, $\int\varphi_s=1$ in view of $\int b\,\mathrm{d}\omega=2$, $\varphi_s\le Cd_s^{-N}$ and $|\varphi_s(y)-\varphi_s(y')|\le C(\max b+\mathrm{Lip}(b))d_s^{-N-1}|y-y'|$ (on its support $|y-x|\ge d_s/2$). In particular, $\alpha(s)\to u(x)$ at a Lebesgue point, the local term is estimated as in the lemma, and the replacement of $\mathcal{J}_b(y-z)$ by $\mathcal{J}_b(x-z)$ in the far term changes the kernel, by (9.1), by at most $C_bd_s|x-z|^{-N-1}$. By the evenness of b, the averages $\bar u_b$ in $\alpha(s)$ and in the far term coincide. Equation (4.3) keeps its form with $\mathcal{V}_b$ instead of $\mathcal{V}$, and $|2\ln(1/d(x))u(x)-\mathcal{V}_b(x)|\le CK$.

Potentials and main estimate. Set $U_{b,x_0}(\xi)=\int_{|y-x_0|>e^{-\xi}}u(y)\mathcal{J}_b(y-x_0)\,\mathrm{d}y$. The proofs of parts (b)–(d) of Lemma 5.1 use the kernel only through two facts. The first is the inequality $||v|^{-N}-|v'|^{-N}|\le N2^{N+1}|v-v'||v|^{-N-1}$ for $|v-v'|\le|v|/2$, comparing two translates of the kernel with centers at distance ρ outside the ball of radius 2ρ around one of the centers, and the second is the mass $2\ln(a'/a)$ of the kernel in the layers $a<|v|<a'$. The same reasoning with (9.1) gives (b)–(d) for U_{b,x_0}, $\mathcal{V}_b$ and the outer potential $\int_\Omega u(x)\mathcal{J}_b(x-y)\,\mathrm{d}x$ with constants depending also on C_b. Part (a) is obtained by passing to polar coordinates: $U'_{b,x_0}(\xi)=\int_{S^{N-1}}b(\omega)u(x_0+e^{-\xi}\omega)\,\mathrm{d}\omega$, and $|U_{b,x_0}(\xi)|\le 2\|u\|_\infty(\xi+\ln D)$ and $|U'_{b,x_0}|\le 2\|u\|_\infty$ (here $b>0$ and $\int b\,\mathrm{d}\omega=2$). In the proof of the main estimate, on the regular part of the sphere $u(y)=U_{b,x_0}(\xi)/(2\xi)+O(K(1+\ln(1/\tau))/\xi)$, and $\int_{\mathcal{S}_r}b\,\mathrm{d}\omega=1+O(r)$, since b is even and the hemispheres $\{\tau>0\}$ and $\{\tau<0\}$ have equal masses with weight b. Therefore estimate (5.4) is preserved in the same form, and the remaining parts of the proof are unchanged. We neither prove nor use below the refinements of Lemma 4.1(b), Lemma 5.1(e), (f) and Remark 5.3 for such operators. Lemma 5.6 is derived only from (1.1) and the estimate $|\beta|\le CK$ and hence is preserved. □

Note that the exponent $-1/2$ in the boundary behavior $u\approx\beta(\ln(1/d))^{-1/2}$ and the very form of the equation $U'=U/(2\xi)$ do not depend on the anisotropy: they are determined only by the fact that a hemisphere carries half the mass of the even kernel. This is how the logarithmic case differs from the fractional one, where the boundary term of the Pohozaev identity for anisotropic operators contains the factor $A(\nu)$, the value of the symbol at the normal [32, Theorem 1.1].

9.2. **Pohozaev identity and Hadamard formula.**

Proof of Theorem 1.6(b), (c). Dilation. Since $b(\theta)|z|^{-N}$ is homogeneous of degree $-N$, equality (6.1) holds for its form. Under dilation and translation the remaining terms become $\langle(V\circ\Phi_h)u,v\rangle$ and $\langle\kappa_h*u,v\rangle$, where $\kappa_h(y)=(1+h)^N\kappa((1+h)y)$ for the dilation and $\kappa_h=\kappa$ for the translation. Hence T_hz is a weak solution in Ω_h of the problem $L(T_hz)=\mu T_hz+f_h$ with

$$f_h=T_hf-2\ln(1+h)T_hz+T_h[(V\circ\Phi_h-V)z+(\kappa_h-\kappa)*z],$$

where the second term is absent for a translation.

Cross identity. For functions with disjoint supports

$$E(\varphi,\psi)=-\iint\varphi(x)\psi(y)(b(\theta)|x-y|^{-N}-\kappa(x-y))\,\mathrm{d}x\,\mathrm{d}y,$$

and the term with V equals zero. The terms with κ on the left-hand side of (6.4) have the form $\int_{\Sigma_\pm}\int_A\kappa(x-y)z_i(y)z_j(x)\,\mathrm{d}x\,\mathrm{d}y$ with $\{i,j\}=\{1,2\}$ and do not exceed $\|\kappa\|_{L^1(B_{2D})}K(\|z_2\mathbb{1}_{\Sigma_+}\|_{L^1}+\|z_1\mathbb{1}_{\Sigma_-}\|_{L^1})=O(h\ell(Ch)^{1/2})=o(h)$ by Theorem 1.6(a) applied to z_1 in Ω and to z_2 in Ω_h. The right-hand side of (6.4) acquires the term $\langle T_h[(V\circ\Phi_h-V)z+(\kappa_h-\kappa)*z],z_1\mathbb{1}_A\rangle$. For $V\in W^{1,1}(U)$ and $\kappa\in W^{1,1}_{\mathrm{loc}}$ its ratio to h tends to $\langle(x\cdot\nabla V)z,z\rangle+\langle(N\kappa+y\cdot\nabla\kappa)*z,z\rangle$ for the dilation and to $\langle(a\cdot\nabla V)z,z\rangle$ for the translation. This uses the convergence of difference quotients in L^1 and the fact that $z_1\mathbb{1}_A$ and T_hz are bounded and

converge to z in L^2. Assembling the terms, as in Section 7.2, gives (1.6), provided Lemma 7.1 holds for the kernel $\mathcal{J}_b$.

Layers. Let w and w_h be weak solutions of the problems for L with bounded right-hand sides, satisfying the remaining assumptions of part (b) of the same lemma with the boundary coefficients of Theorem 1.6(a). The assumptions of part (a) of that lemma hold with $\mathcal{J} = \mathcal{J}_b$, $M = \max b$, $u = w$, $v = w_h$ and $\alpha = \gamma = \beta(w)$. Indeed, the potential $\mathcal{P}_b(y) = \int_\Omega w(x)\mathcal{J}_b(x-y)\,\mathrm{d}x$ for $y \in \Sigma_+$ equals $2\beta(y_0)(\ln(1/\sigma))^{1/2} + O(K)$ by Lemma 5.1(d), (b) for $\mathcal{J}_b$ and the estimates (5.5) for U_b and (1.2), that is, it has the same form as for $b \equiv c_N$, and the integral of w_h outside Ω_h is estimated in the same way. The values of w and w_h in the layers are given by the same theorem in Ω and in Ω_h with constants common to all h. Therefore part (b) of the layer lemma and Lemma 8.7 hold with $\mathcal{J}_b(x-y)$ instead of $c_N|x-y|^{-N}$. The assumptions of part (b) of the layer lemma for $w = z$ and $w_h = T_h z$ hold, since the right-hand side f_h is bounded uniformly in h (the norms of $\kappa \mathbb{1}_{B_{2D}}$ and $\kappa_h \mathbb{1}_{B_{2D}}$ in L^1 are finite with a bound independent of h), and $\beta(T_h z)(\Phi_h(x_0)) = J_h^{-1/2}\beta(z)(x_0)$, as in Section 7.

Form transport. For the form (1.5) in Lemma 8.1 the difference $E(S_t u, S_t v) - E(u,v)$ is estimated by the quantity $\|u\|_H \|v\|_H$ multiplied by a factor that does not depend on u, v and tends to zero as $t \to 0$. In the local and far terms the weight additionally contains the ratio $b(\theta_t)/b(\theta)$, where θ_t is the direction of $\Phi_t(x') - \Phi_t(y')$. Since $|\theta_t - \theta| \le 2|t|\|\nabla X\|_\infty$, this ratio equals $1 + O(t)$ with a constant depending on $\operatorname{Lip}(b)/\min b$, and these terms change, as in the lemma, by $O(t)\|u\|_H\|v\|_H$. For κ the kernel $\kappa(\Phi_t(x) - \Phi_t(y))J_t(x)J_t(y) - \kappa(x-y)$ tends to zero in the sense of $\sup_{x\in\Omega}\int_\Omega |\dots|\,\mathrm{d}y \to 0$, which can be seen by approximating κ by continuous functions in $L^1(B_{2D})$, so that by the Schur test the norm of the corresponding operator in L^2 is small. For V the function $W_t = (V \circ \Phi_t)J_t - V$ is bounded and tends to zero in L^1, so $\|W_t\|_{L^2}^2 \le \|W_t\|_\infty \|W_t\|_{L^1} \to 0$. The unit ball of $H(\Omega)$ is precompact in L^2 [6, (3.3)] and is covered by finitely many L^2-balls of radius ε with centers $u_i \in L^\infty$. If $\|u - u_i\| \le \varepsilon$, then $|\int W_t uv| \le \|W_t\|_\infty \varepsilon \|v\| + \|u_i\|_\infty \|W_t\|_{L^2}\|v\|$, and therefore $\sup|\int W_t uv| \to 0$, where the supremum is taken over u, v in the unit ball of $H(\Omega)$.

Eigenvalue continuity. The proof of Lemma 8.2 goes through with this factor in place of $C|t|$. The inequality $E(u,u) + C\|u\|^2 \ge c\|u\|_H^2$ is preserved, and the minimax gives $\Lambda_k(\Omega_t) \to \Lambda_k(\Omega)$ for each k. The argument about the eigenfunctions is unchanged: the estimate $\|\varphi_t\|_\infty \le C$ follows from Section 9.1, whose constants are common to all Ω_t with $\operatorname{diam}\Omega_t \le 2D$. In the part about the boundary coefficients the kernel $\mathbb{1}_{|y-x|>e^{-\xi}}\mathcal{J}_b(y-x)$ is at most $\max b\, e^{N\xi}$ and depends continuously on x in L^1_{loc}. We obtain all the assertions of the same lemma, except $|\Lambda(t) - \Lambda| \le C|t|$, for which only $\Lambda(t) \to \Lambda$ is known so far.

Hadamard formula. The terms with V in (8.1) cancel exactly, since V is the same in both domains, and the terms with κ are $o(t)$, as above. By the layer lemma for the kernel $\mathcal{J}_b$ the right-hand side of (8.1) equals $-2t\int \beta^2 (X\cdot\nu)\,\mathrm{d}\sigma + o(t)$, while $\langle\varphi_t, \varphi \mathbb{1}_A\rangle \to 1$, and the proof of Theorem 1.4 goes through without changes. Hence one also obtains the rate: the right-hand side of (8.1) is $O(|t|)$, and $|\Lambda(t) - \Lambda| \le C|t|$ for small $|t|$. □

Theorem 8.8 for L is proved in the same way, with one change: the estimate $|\Lambda^{(k)}(t) - \Lambda| \le C|t|$ follows from the cross identity. Indeed, as in the part "Eigenfunctions" of the proof of Lemma 8.2, applied to ψ_k^t, the matrix $(\langle\psi_k^t, \varphi_i \mathbb{1}_A\rangle)$ is close to orthogonal. Therefore, for each k there exists i with $|\langle\psi_k^t, \varphi_i\mathbb{1}_A\rangle| \ge 1/(2\sqrt{m})$, and the right-hand side of the identity is $O(t)$.

9.3. **The logarithmic Schrödinger operator.** The proof of Corollary 1.7 relies on the following lemma.

Lemma 9.1 (kernel of $(I-\Delta)^{\log}$)**.** *The kernel of the operator $(I-\Delta)^{\log}$ equals*

$$j(y) = c_N \frac{\omega_N(|y|)}{\omega_N(0)}|y|^{-N}, \quad \omega_N(r) = 2^{1-N/2} r^{N/2} K_{N/2}(r), \quad \omega_N(0) = \Gamma(N/2),$$

where K_ν is the modified Bessel function of the second kind. The function $\kappa = c_N|y|^{-N} - j$ is positive, even and lies in $W^{1,1}_{\mathrm{loc}}(\mathbb{R}^N)$. For $N \ge 2$ and $0 < r = |y| \le 1$, $|\kappa(y)| + r|\nabla\kappa(y)| \le Cr^{2-N}(1 + |\ln r|)$ holds, and for $N = 1$ we have $\kappa(y) = (1 - e^{-|y|})/|y|$ and $|\kappa'| \le 1/2$. The form E_{J} of the operator $(I-\Delta)^{\log}$ equals $E_{\mathrm{L}} + \langle\kappa * \cdot, \cdot\rangle$, and the space from [14] coincides with $H(\Omega)$ with an equivalent norm.

Proof. The formula for j is [14, Theorem 1.1]. By the identity $(r^{N/2}K_{N/2})' = -r^{N/2}K_{N/2-1}$ [27, (10.29.4)], we have $\omega_N'(r) = -2^{1-N/2}r^{N/2}K_{N/2-1}(r) < 0$, so $j < c_N|y|^{-N}$ and $\kappa > 0$. The kernel j is integrable at infinity since $K_{N/2}(r)$ decays exponentially as $r \to \infty$ [27, (10.25.3)]. For $N \ge 2$ and $r \le 1$, the formula for ω_N' and the behavior of $K_{N/2-1}$ at zero [27, (10.30.2), (10.30.3)] imply $|\omega_N'(r)| \le Cr(1 + |\ln r|)$ and $\omega_N(0) - \omega_N(r) \le Cr^2(1 + |\ln r|)$, which gives the estimate for κ, while for $N = 1$ we have $\omega_1(r) = \sqrt{\pi}e^{-r}$ [27, (10.39.2)] and $\kappa(y) = (1 - e^{-|y|})/|y|$, whence $|\kappa'| \le 1/2$, since $1 - e^{-y}(1+y) \le y^2/2$ for $y > 0$.

Expanding $(1/2)\iint(u(x)-u(y))(v(x)-v(y))j(x-y)\,\mathrm{d}x\,\mathrm{d}y$ in the same way as in the proof of Lemma 3.1, we obtain form (1.5) with $b\equiv c_N$, $\kappa=c_N|y|^{-N}-j$ and a constant $V=\int_{|z|\ge1}j\,\mathrm{d}z-\int_{|z|<1}\kappa\,\mathrm{d}z$. This constant equals ρ_N: the difference of the symbols $\ln(1+|k|^2)-2\ln|k|=\ln(1+|k|^{-2})$ equals $V-\rho_N+\hat\kappa(k)$, and $\hat\kappa(k)\to0$ as $|k|\to\infty$, since $\kappa\mathbb{1}_{|z|<1}\in L^1$, and $\kappa\mathbb{1}_{|z|\ge1}$ is the sum of an integrable function and a function $c_N|z|^{-N}\chi(z)$, where χ is a smooth function equal to zero for $|z|\le1/2$ and to one for $|z|\ge1$. The gradient of the function $c_N|z|^{-N}\chi(z)$ is integrable, so the Fourier transform of this function is $O(1/|k|)$ as $|k|\to\infty$. Hence $E_{\mathrm{J}}=E_{\mathrm{L}}+\langle\kappa*\cdot,\cdot\rangle$. Finally, $(\omega_N(1)/\omega_N(0))c_N|y|^{-N}\le j(y)\le c_N|y|^{-N}$ for $|y|<1$, and j is integrable for $|y|\ge1$, which gives the equivalence of the norms. □

Proof of Corollary 1.7. By Lemma 9.1 the form of $(I-\Delta)^{\log}$ is of type (1.5) with $b\equiv c_N$, $V\equiv\rho_N$ and an even $\kappa\in W^{1,1}_{\mathrm{loc}}$, and conditions (B1)–(B3) hold.

Here $t^N|ty|^{-N}$ does not depend on t for $t>0$, and therefore, with $r=|y|$,

$$N\kappa+y\cdot\nabla\kappa=-\frac{\mathrm{d}}{\mathrm{d}t}t^Nj(ty)|_{t=1}=-\frac{c_Nr\omega_N'(r)}{\omega_N(0)r^N}.$$

By $(r^{N/2}K_{N/2})'=-r^{N/2}K_{N/2-1}$

$$N\kappa+y\cdot\nabla\kappa=2(2\pi)^{-N/2}r^{1-N/2}K_{N/2-1}(r)=2G(y),$$

where G is the Bessel kernel of $(I-\Delta)^{-1}$. Substituting this into (1.6) with $\nabla V=0$, we obtain (1.7). Finally, $\|z\|^2-\langle G*z,z\rangle=(2\pi)^{-N}\int|k|^2(1+|k|^2)^{-1}|\hat z(k)|^2\,\mathrm{d}k$, and for $z\ne0$ this quantity is positive and less than $\|z\|^2$. □

Corollary 9.2 (simplicity of the spectrum of $(I-\Delta)^{\log}$ on an interval)**.** *All eigenvalues of the Dirichlet problem for $(I-\Delta)^{\log}$ on an interval are simple.*

Proof. Let φ be a normalized real eigenfunction on (p,q). The equality (1.7) with $a=p$ and $f=0$ gives $(q-p)\beta_q(\varphi)^2=1-\langle G*\varphi,\varphi\rangle>0$. If φ_1, φ_2 are orthonormal eigenfunctions with the same eigenvalue and $\alpha_i=\beta_q(\varphi_i)\ne0$, then for $\psi=\alpha_2\varphi_1-\alpha_1\varphi_2$ we have $\beta_q(\psi)=0$, and this equality gives $\|\psi\|^2-\langle G*\psi,\psi\rangle=0$, which is impossible, since $\psi\ne0$ by the orthogonality of φ_1 and φ_2. □

This corollary also follows from [22, Theorem 1.2], since $t\mapsto\ln(1+t)$ is a complete Bernstein function that does not extend meromorphically to $\mathbb{C}\setminus\{0\}$. The proof given here additionally yields $\beta_q(\varphi)\ne0$ for each eigenfunction φ.

Corollary 9.3 (monotone potential)**.** *Let $N=1$, $V\in L^\infty(\mathbb{R})$, $V\in W^{1,1}(U)$ for a neighborhood U of the closed interval $[p,q]$ and $V'\le0$ almost everywhere on (p,q) or $V'\ge0$ almost everywhere on (p,q). Then all eigenvalues of the Dirichlet problem on (p,q) for the operators $L_\Delta+V$ and $(I-\Delta)^{\log}+V$ are simple.*

Proof. Here $L_\Delta+V$ has the form (1.5) with $b\equiv c_1$, $\kappa=0$ and potential ρ_1+V, and the operator $(I-\Delta)^{\log}+V$, with the same b, the potential ρ_1+V and the kernel κ of Lemma 9.1. The neighborhood U may be assumed bounded, and in both cases the assumptions of Theorem 1.6(b) hold, for the second operator by Corollary 1.7, and the gradient of the potential equals V'. Let $V'\le0$ almost everywhere on (p,q) and $a=p$. For a real eigenfunction φ, the equality (1.6) with $f=0$ gives

$$(q-p)\beta_q(\varphi)^2=\|\varphi\|^2-\frac12\int_p^q(x-p)V'(x)\varphi(x)^2\,\mathrm{d}x-\frac12\langle(\kappa+y\kappa')*\varphi,\varphi\rangle.$$

The integral is nonpositive. The last term vanishes for $L_\Delta+V$ and equals $-\langle G*\varphi,\varphi\rangle$ for $(I-\Delta)^{\log}+V$ (the proof of the same corollary), so the right-hand side is at least $\|\varphi\|^2-\langle G*\varphi,\varphi\rangle>0$ for $\varphi\ne0$. If φ_1, φ_2 are orthonormal eigenfunctions of the same eigenvalue and $\alpha_i=\beta_q(\varphi_i)$, then the eigenfunction $\psi=\alpha_2\varphi_1-\alpha_1\varphi_2$ has $\beta_q(\psi)=0$ by the linearity of β, and this inequality gives $\psi=0$. But $\alpha_i\ne0$, and $\|\psi\|^2=\alpha_1^2+\alpha_2^2>0$. For $V'\ge0$ we take $a=q$. Then $(x-q)V'\le0$ on (p,q), and the argument is repeated with β_p instead of β_q. □

For $V\in W^{1,1}(U)$ the condition on the sign of V' means that V is monotone on $[p,q]$.

Corollary 9.4 (the ball, the annulus and the union of intervals)**.** *Corollaries 7.5 and 7.7 are valid for $(I-\Delta)^{\log}$.*

Proof. Their proofs use only two facts. The first is the boundary expansion of Theorem 1.6(a), whose derivation, as in Sections 3–7, does not use the connectedness of Ω. The second is the positivity of the right-hand side of the Pohozaev identity of Theorem 1.6(b), whose derivation does not use connectedness either, for an eigenfunction $\varphi\ne0$, which here equals $\|\varphi\|^2-\langle G*\varphi,\varphi\rangle>0$. □

Proof of Theorem 8.11 for $(I-\Delta)^{\log}$. For $(I-\Delta)^{\log}$ the sets $\mathcal{X}_n$, defined as in the proof for L_Δ by the eigenvalues of $(I-\Delta)^{\log}$, are open, since the eigenvalues depend continuously on X: by Section 9.2 the change of the term with κ is small by the Schur test uniformly for small $\|\Phi-\mathrm{id}\|_{C^1}$, where Φ is the map from the proof for L_Δ. Theorem 8.8 for this operator holds by Section 9.2, and in the proof of Lemma 8.10, Corollary 1.3 is replaced by the last statement of Corollary 1.7, according to which the boundary coefficient of an eigenfunction does not vanish identically. □

In contrast to L_Δ, for $(I-\Delta)^{\log}$ the right-hand side of (1.7) depends on the eigenfunction itself, and β^2 for a normalized radial eigenfunction of the ball equals $(1-\langle G*\varphi,\varphi\rangle)/(N|B_R|)$. For a simple eigenvalue and a normalized eigenfunction, under a dilation the Hadamard formula gives $\mathrm{d}\Lambda(t\Omega)/\,\mathrm{d}t|_{t=1} = -2(1-\langle G*\varphi,\varphi\rangle)$, which coincides with the derivative of the Rayleigh quotient computed via the symbol $\ln(1+|k|^2)$.

10. Comparison with the one-dimensional theory

We check the results against the one-dimensional theory on an interval and against radial reduction in dimension three. Neither check is used in the proofs: they verify the constant factors in the identity (1.3) and in Corollary 1.3 by independent computations.

10.1. **Interval.** Let $N=1$ and $\Omega=(p,q)$. The boundary consists of the points p and q, with $\nu(p)=-1$ and $\nu(q)=1$. Then (1.3) for $a=p$ takes the form

$$(q-p)\beta_q^2 = \|z\|^2 + \frac{1}{2}\langle f,z\rangle + \langle z,(x-p)f'\rangle,$$

that is, the first equality of [4, Theorem 4.8] divided by 2, and the second is obtained for $a=q$. For a normalized eigenfunction on $(-R,R)$ Corollary 1.3 for a equal to the left and right endpoints gives $2R\beta_+^2 = 2R\beta_-^2 = 1$ for the boundary coefficients $\beta_\pm$ at the points $\pm R$, that is, [4, Corollary 4.9], with $1/(2R) = 1/(N|B_R|)$ in agreement with the formula of the same corollary for the ball.

10.2. **Radial functions in $\mathbb{R}^3$.** In dimension three, radial functions reduce to odd functions on the line, and the formula of Corollary 1.3 for the ball is obtained from the one-dimensional one.

Proposition 10.1 (radial reduction for $N=3$)**.** *Let u be a radial function on $\mathbb{R}^3$ and $w(r)=ru(|r|)$, $r\in\mathbb{R}$. If u is smooth and has compact support, then $r(L_\Delta u)(r) = (L_\Delta w)(r)$, where the operator on the left acts in $\mathbb{R}^3$ and the one on the right in $\mathbb{R}$. If u is a radial eigenfunction of $B_R\subset\mathbb{R}^3$ with eigenvalue Λ, then w is an odd eigenfunction of $(-R,R)$ with the same Λ, and $\|u\|^2 = 2\pi\|w\|^2$.*

Proof. The Fourier transforms of a radial function u and an odd function w have the form

$$\hat u(k) = \frac{4\pi}{|k|}\int_0^\infty ru(r)\sin(|k|r)\,\mathrm{d}r, \quad \hat w(k) = -2i\int_0^\infty w(r)\sin(kr)\,\mathrm{d}r.$$

Hence, $\hat u(k) = 2\pi i\hat w(|k|)/|k|$, and since both operators multiply the Fourier transform by $2\ln|k|$ (the relations are understood in L^2), the first assertion follows from the same relation for $L_\Delta u$ and $L_\Delta w$. By this formula $E_{\mathrm{L}}(u,v) = 2\pi E_{\mathrm{L}}(w,w_v)$ and $\langle u,v\rangle = 2\pi\langle w,w_v\rangle$ for smooth radial u, v supported in B_R, where $w_v(r)=rv(|r|)$. A direct check gives $\|u\|^2 = 4\pi\int_0^R r^2u^2\,\mathrm{d}r = 2\pi\|w\|^2$. Averaging over rotations maps smooth functions to smooth radial ones, so by [6, Theorem 3.1] smooth radial functions are dense in the radial part of $H(B_R)$, and both identities extend by continuity. Conversely, a smooth odd function ψ with compact support in $(-R,R)$ is w_v for the smooth radial $v(x)=\psi(|x|)/|x|$, $v(0)=\psi'(0)$, and smooth odd functions are dense in the odd part of $H((-R,R))$. Hence the weak equation for u on radial test functions is equivalent to the weak equation for w on odd test functions, and on even ones it holds since both sides are equal to zero: the form is invariant under reflection, and for an even test function v we have $E_{\mathrm{L}}(w,v) = -E_{\mathrm{L}}(w,v)$. □

If u is a normalized radial eigenfunction of $B_R\subset\mathbb{R}^3$, then $\sqrt{2\pi}w$ is normalized, and by [4, Corollary 4.9] $\beta(w)^2 = 1/(4\pi R)$. Since $u(x) = w(R-\delta)/(R-\delta)$ for $|x| = R-\delta$, the boundary coefficient of u equals $\beta(w)(R)/R$, and

$$\beta(u)^2 = \frac{1}{4\pi R^3} = \frac{1}{3|B_R|},$$

which coincides with Corollary 1.3. This derivation uses only the one-dimensional theorem and Proposition 10.1 and does not depend on Section 7.

11. Conclusion

We have proved that every weak solution of the Dirichlet problem for the logarithmic Laplacian with a bounded right-hand side in a domain of class $C^{1,1}$ has a continuous boundary coefficient β, and found the rate of convergence and the modulus of continuity. From this follow the Pohozaev identity for every μ and the Hadamard formula for simple and multiple eigenvalues and, when zero is not an eigenvalue, for the torsional rigidity. The Hadamard formula yields the generic simplicity of the spectrum and, for $N \geq 2$, the strict decrease of the principal eigenvalue of a ball with an off-center hole as the hole moves away from the center and, if $\Lambda_1(B_R) > 0$, the strict increase of the torsional rigidity of this domain. The boundary expansion, the Pohozaev identity with additional terms (under the regularity assumptions of Theorem 1.6(b)) and the Hadamard formula for simple and multiple eigenvalues carry over to operators of logarithmic order, and for $(I-\Delta)^{\log}$ the generic simplicity of the spectrum is also proved. On an interval, the simplicity of the spectrum of the operators $L_\Delta + V$ and $(I-\Delta)^{\log} + V$ with a monotone potential of class $W^{1,1}$ is also established, and for $L_\Delta + V$ this generalizes the simplicity of the spectrum of L_Δ on an interval [4].

No a priori regularity of the solution near the boundary is used. The boundary expansion is derived from a first-order linear equation for potentials centered on the boundary, in which the interaction between boundary points enters only through the bounded right-hand side. The Pohozaev identity and the Hadamard formula are obtained from an exact cross identity for two solutions in nearby domains. The exponent $-1/2$ and the form of the boundary term do not depend on the anisotropy of the operator: both are determined by the fact that each hemisphere carries half the mass of the even kernel.

Limitations. The boundary expansion, the Pohozaev identity and the Hadamard formula are proved for bounded domains of class $C^{1,1}$. The inhomogeneous term f in Theorem 1.2 is assumed to lie in $L^\infty(\Omega) \cap W^{1,1}(\Omega)$. For a bounded solution of a semilinear equation with a continuous nonlinearity the first condition holds, while the second requires differentiability of the nonlinearity in u and interior regularity of the solution, which is not proved in this paper.

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